

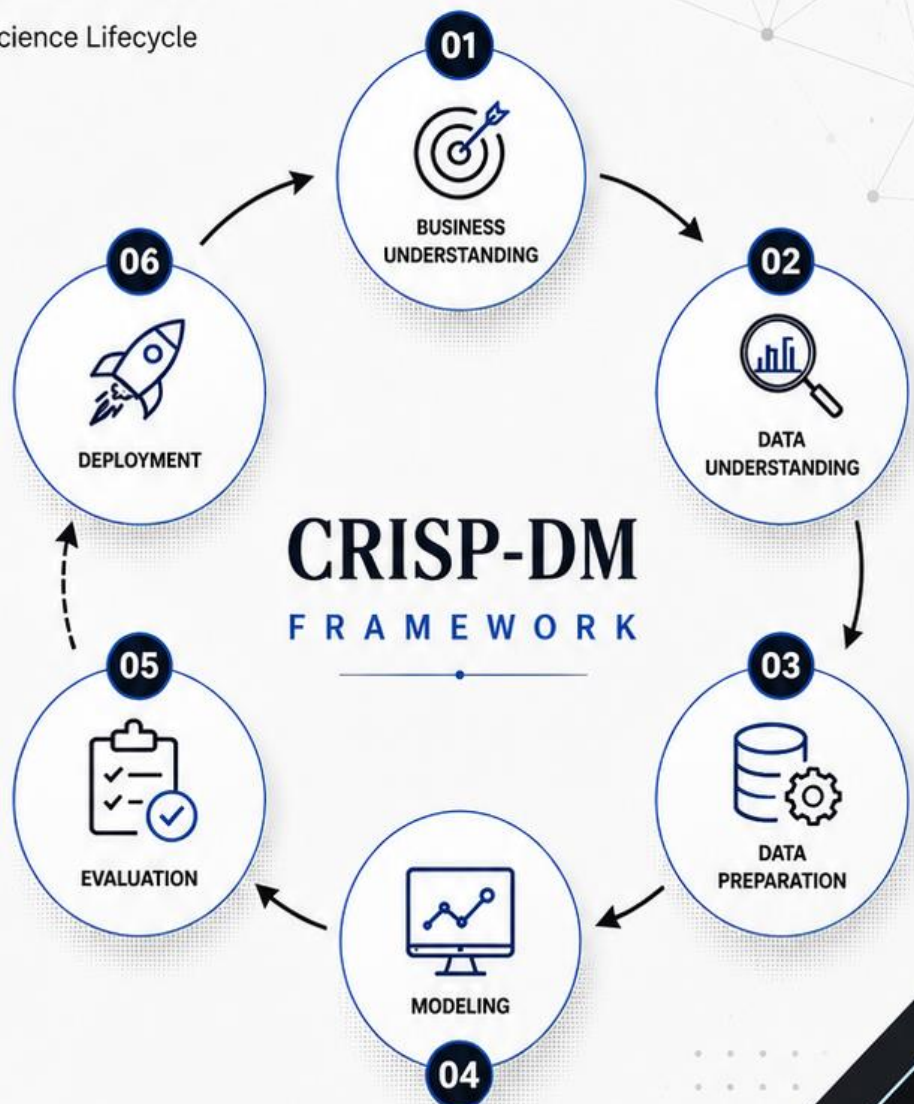
DATA SCIENCE METHODOLOGY

COMPREHENSIVE LEARNING NOTES & PROJECT REPORT

Understanding the End-to-End Data Science Lifecycle
using the **CRISP-DM** Framework

TOPICS COVERED

- Data Science Methodology
- CRISP-DM Framework
- Business Understanding
- Data Collection
- Data Preparation
- Data Preprocessing
- Predictive Analytics
- Descriptive Analytics
- Classification Models
- Model Training
- Model Evaluation
- Model Deployment
- Data Storytelling
- Jupyter Notebook
- Python



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Table of Contents

1	<u>DATA SCIENCE METHODOLOGY OVERVIEW</u>	1
1.1	LEARNING OBJECTIVES	1
1.2	INTRODUCTION	1
1.3	WHAT IS A METHODOLOGY?	1
1.4	DATA SCIENCE METHODOLOGY	2
1.5	THE 10 STAGES OF DATA SCIENCE METHODOLOGY	2
1.6	THE IMPORTANCE OF ASKING QUESTIONS	2
1.6.1	DEFINING THE BUSINESS PROBLEM AND APPROACH	2
1.6.2	ORGANIZING THE DATA	2
1.6.3	VALIDATING THE SOLUTION	3
1.7	KEY TAKEAWAYS	3
2	<u>DATA SCIENCE METHODOLOGY 101: FROM PROBLEM TO APPROACH – BUSINESS UNDERSTANDING</u>	3
2.1	INTRODUCTION	3
2.2	WHY BUSINESS UNDERSTANDING IS IMPORTANT	4
2.3	DEFINING THE BUSINESS PROBLEM	4
2.4	UNDERSTANDING THE GOAL BEFORE THE QUESTION	4
2.5	DEFINING OBJECTIVES	5
2.6	CASE STUDY: HEALTHCARE BUDGET ALLOCATION	5
2.6.1	BUSINESS PROBLEM	5
2.7	ESTABLISHING GOALS AND OBJECTIVES	5
2.8	KEY FINDINGS	6
2.9	SELECTING THE ANALYTICAL APPROACH	6
2.10	IBM DATA SCIENTISTS' APPROACH	6
2.11	ROLE OF THE BUSINESS SPONSOR	6
2.12	BUSINESS REQUIREMENTS	6
2.13	KEY TAKEAWAYS	7
3	<u>DATA SCIENCE METHODOLOGY 101: FROM PROBLEM TO APPROACH (ANALYTIC APPROACH)</u>	7
3.1	OVERVIEW	7
3.2	SELECTING THE APPROPRIATE ANALYTIC APPROACH	7
3.3	MACHINE LEARNING IN THE ANALYTIC APPROACH	8
3.4	CASE STUDY: APPLYING THE ANALYTIC APPROACH	8
3.5	WHY A DECISION TREE CLASSIFICATION MODEL WAS CHOSEN	9
3.6	KEY TAKEAWAYS	9

4	<u>ANALYTIC APPROACH BASED ON THE QUESTION TYPE</u>	9
4.1	DESCRIPTIVE QUESTIONS: "WHAT IS THE CURRENT STATUS?"	9
4.2	DIAGNOSTIC QUESTIONS: "WHY DID IT HAPPEN?"	10
4.3	PREDICTIVE QUESTIONS: "WHAT IS LIKELY TO HAPPEN?"	10
4.4	PRESCRIPTIVE QUESTIONS: "WHAT SHOULD WE DO?"	11
4.5	CLASSIFICATION QUESTIONS: "WHICH CATEGORY DOES THIS BELONG TO?"	11
4.6	SUMMARY	12
5	<u>BUSINESS UNDERSTANDING: IDENTIFYING RELEVANT QUESTIONS</u>	12
5.1	OVERVIEW	12
5.2	RELEVANT QUESTIONS TO THE BUSINESS GOAL	12
5.3	QUESTIONS THAT ARE NOT RELEVANT TO THE BUSINESS GOAL	13
6	<u>ANALYTICAL APPROACH: IDENTIFYING THE PATTERN TO ADDRESS THE QUESTION</u>	13
6.1	OVERVIEW	13
6.2	PREDICTIVE MODEL	13
6.3	DESCRIPTIVE MODEL	13
6.4	CLASSIFICATION MODEL	14
6.5	CLUSTERING (GROUPING-BASED ANALYTICAL APPROACH)	14
7	<u>FROM PROBLEM TO APPROACH</u>	14
7.1	OVERVIEW	14
7.2	BUSINESS UNDERSTANDING: UNDERSTANDING THE QUESTION	14
7.3	ANALYTICAL APPROACH: DETERMINE THE APPROPRIATE APPROACH	15
7.4	KEY TAKEAWAY	15
8	<u>GLOSSARY: FROM PROBLEM TO APPROACH</u>	15
8.1	OVERVIEW	15
8.2	GLOSSARY	15
9	<u>DATA SCIENCE METHODOLOGY: DATA REQUIREMENTS</u>	17
9.1	INTRODUCTION	17
9.2	DEFINING DATA REQUIREMENTS	18
9.3	CASE STUDY: APPLYING DATA REQUIREMENTS	18
9.3.1	PATIENT COHORT SELECTION CRITERIA	18
9.4	DATA CONTENT AND FORMAT REQUIREMENTS	19
9.5	PREPARING DATA FOR MODELING	19

10	DATA COLLECTION	20
10.1	ASSESSING THE COLLECTED DATA	20
10.2	CASE STUDY: APPLYING DATA COLLECTION	20
10.3	HANDLING UNAVAILABLE DATA	21
10.4	DATA INTEGRATION	21
10.5	IMPROVING THE DATA COLLECTION PROCESS	21
11	FROM REQUIREMENTS TO COLLECTION	21
11.1	OVERVIEW	21
11.2	DATA REQUIREMENTS AND COLLECTION: ENSURING THE RIGHT INGREDIENTS	22
11.2.1	DATA REQUIREMENTS	22
11.2.2	DATA COLLECTION	22
11.2.3	DATA INTEGRATION	22
11.2.4	DATA UNDERSTANDING	22
11.3	KEY TAKEAWAY	23
12	GLOSSARY: FROM REQUIREMENTS TO COLLECTION	23
12.1	OVERVIEW	23
12.2	GLOSSARY	23
13	DATA SCIENCE METHODOLOGY 101: DATA UNDERSTANDING	24
13.1	INTRODUCTION	24
13.2	DATA UNDERSTANDING IN THE CONGESTIVE HEART FAILURE CASE STUDY	24
13.3	PAIRWISE CORRELATION ANALYSIS	24
13.4	USING HISTOGRAMS TO UNDERSTAND DATA DISTRIBUTION	25
13.5	ASSESSING DATA QUALITY	25
13.6	REFINING THE DEFINITION OF CONGESTIVE HEART FAILURE ADMISSION	26
13.7	THE ITERATIVE NATURE OF DATA SCIENCE METHODOLOGY	26
14	DATA SCIENCE METHODOLOGY 101: DATA PREPARATION – CONCEPTS	26
14.1	INTRODUCTION	26
14.2	IMPORTANCE OF DATA PREPARATION	26
14.3	TRANSFORMING DATA	27
14.4	PREPARING DATA FOR ANALYSIS	27
14.5	FEATURE ENGINEERING	27
14.6	TEXT ANALYSIS DURING DATA PREPARATION	27
14.7	IMPORTANCE OF COMPLETING DATA PREPARATION CAREFULLY	28
14.8	KEY TAKEAWAY	28

15	DATA PREPARATION – CASE STUDY	28
15.1	DEFINING CONGESTIVE HEART FAILURE	28
15.2	DEFINING READMISSION CRITERIA	28
15.3	AGGREGATING TRANSACTIONAL DATA	29
15.4	CREATING PATIENT-LEVEL FEATURES	29
15.5	CONSIDERING CO-MORBIDITIES	30
15.6	LITERATURE REVIEW AND DATA COLLECTION REFINEMENT	30
15.7	MERGING PATIENT INFORMATION	30
15.8	TARGET VARIABLE	31
15.9	FINAL PREPARED DATASET	31
16	LESSON SUMMARY: FROM UNDERSTANDING TO PREPARATION	31
16.1	OVERVIEW	31
16.2	DATA UNDERSTANDING AND PREPARATION	32
16.2.1	DATA UNDERSTANDING AND ITERATIVE ASSESSMENT	32
16.2.2	IMPORTANCE OF DATA UNDERSTANDING	32
16.2.3	ASSESSMENT AND ITERATION	32
16.3	DATA PREPARATION	32
16.4	EFFICIENCY AND QUALITY	32
16.5	KEY TAKEAWAYS	32
17	MODULE 2 LESSON 1 GLOSSARY: FROM UNDERSTANDING TO PREPARATION	33
17.1	OVERVIEW	33
17.2	GLOSSARY	33
18	DATA SCIENCE METHODOLOGY: MODELING – CONCEPTS	34
18.1	INTRODUCTION	34
18.2	PURPOSE OF DATA MODELING	35
18.2.1	DESCRIPTIVE MODELS	35
18.2.2	PREDICTIVE MODELS	35
18.3	TRAINING SET FOR PREDICTIVE MODELING	35
18.4	IMPORTANCE OF DATA PREPARATION AND MODELING	35
18.5	JOHN ROLLINS' DESCRIPTIVE DATA SCIENCE METHODOLOGY	36
18.6	MOVING TOWARD MODEL EVALUATION	36
18.7	IMPORTANCE OF RELEVANCE IN DATA SCIENCE	36
18.8	SUMMARY	36
19	MODELING – CASE STUDY (DATA SCIENCE METHODOLOGY)	37

19.1	INTRODUCTION	37
19.2	BUILDING THE INITIAL MODEL	37
19.2.1	RESULTS OF THE FIRST MODEL	37
19.3	IMPROVING THE MODEL THROUGH PARAMETER TUNING	37
19.3.1	TYPE I ERROR (FALSE POSITIVE)	38
19.3.2	TYPE II ERROR (FALSE NEGATIVE)	38
19.4	DEFAULT RELATIVE COST	38
19.5	SECOND MODEL	38
19.5.1	RESULTS	39
19.6	THIRD MODEL	39
19.6.1	RESULTS	39
19.6.2	PERFORMANCE SUMMARY	39
19.7	ITERATIVE NATURE OF MODELING	40
19.8	KEY TAKEAWAYS	40
20	MODEL EVALUATION	40
20.1	PHASES OF MODEL EVALUATION	41
20.1.1	DIAGNOSTIC MEASURES	41
20.1.2	STATISTICAL SIGNIFICANCE TESTING	41
20.2	CASE STUDY: FINDING THE OPTIMAL MODEL	42
20.3	BUSINESS CONSIDERATIONS	42
20.4	RECEIVER OPERATING CHARACTERISTIC (ROC) CURVE	42
20.4.1	WHAT IS THE ROC CURVE?	43
20.4.2	PURPOSE OF THE ROC CURVE	43
20.5	KEY TAKEAWAYS	43
21	LESSON SUMMARY: FROM MODELING TO EVALUATION	44
21.1	MODELING TO EVALUATION	44
21.1.1	DATA MODELING	44
21.1.2	TRAINING SET AND CALIBRATION	44
21.1.3	MODEL EVALUATION	44
21.1.4	DIAGNOSTIC MEASURES	45
21.2	KEY TAKEAWAYS	45
22	GLOSSARY: FROM MODELING TO EVALUATION	45
22.1	OVERVIEW	45
22.2	GLOSSARY	45
23	DEPLOYMENT	47

23.1	CASE STUDY: DEPLOYMENT FOR REDUCING HOSPITAL READMISSION RISK	47
23.2	BUSINESS REQUIREMENTS	47
23.3	PATIENT RISK ASSESSMENT PROCESS	48
23.4	TRAINING AND OPERATIONAL SUPPORT	48
23.5	EXAMPLE: COGNOS APPLICATION	48
23.6	INTERACTIVE SUMMARY REPORT	49
23.7	INDIVIDUAL PATIENT REPORT	49
23.8	KEY TAKEAWAYS	49
24	FEEDBACK	50
24.1	INTRODUCTION	50
24.2	PURPOSE OF THE FEEDBACK STAGE	50
24.3	FEEDBACK STAGE PLAN	50
24.4	MEASURING MODEL EFFECTIVENESS	50
24.5	MODEL REFINEMENT	51
24.6	REFINEMENT OF INTERVENTION ACTIONS	51
24.7	KEY TAKEAWAYS	51
25	THE ROLE OF STORYTELLING IN THE LIFE OF A DATA ANALYST	52
25.1	WHY STORYTELLING MATTERS	52
25.2	BALANCING SIMPLICITY AND COMPLEXITY	52
25.3	THE IMPORTANCE OF COMMUNICATION	52
25.4	STORYTELLING AS THE FINAL STEP	52
25.5	STORYTELLING HELPS DRIVE ACTION	53
25.6	STANFORD STUDY ON STORYTELLING	53
25.7	CREATING AN EMOTIONAL CONNECTION	53
26	DATA SCIENCE METHODOLOGY 101 COURSE SUMMARY	54
26.1	WHAT WE LEARNED	54
26.2	THE ITERATIVE NATURE OF THE DATA SCIENCE METHODOLOGY	55
26.3	APPLYING THE METHODOLOGY IN A REAL-WORLD CASE STUDY	55
26.4	THE TRUE PURPOSE OF A METHODOLOGY	55
26.5	FINAL TAKEAWAY	55
27	MODULE 3 SUMMARY: FROM DEPLOYMENT TO FEEDBACK	56
27.1	FROM DEPLOYMENT TO FEEDBACK	56
27.1.1	STAKEHOLDER ENGAGEMENT	56
27.1.2	DEPLOYMENT AND FEEDBACK	56
27.1.3	ITERATIVE PROCESS	56

27.1.4	IMPROVEMENT AND REDEPLOYMENT	57
28	MODULE 3 GLOSSARY: FROM DEPLOYMENT TO FEEDBACK	57
28.1	GLOSSARY	57
29	INTRODUCTION TO CRISP-DM	58
29.1	LEARNING OBJECTIVES	58
29.2	WHAT IS CRISP-DM?	58
29.3	THE SIX STAGES OF CRISP-DM	59
29.3.1	BUSINESS UNDERSTANDING	59
29.3.2	DATA UNDERSTANDING	59
29.3.3	DATA PREPARATION	60
29.3.4	MODELING	60
29.3.5	EVALUATION	60
29.3.6	DEPLOYMENT	61
29.4	THE ITERATIVE NATURE OF CRISP-DM	61
29.5	WHAT HAPPENS AFTER THE FINAL STAGE?	61
29.6	SUMMARY	61
30	FINAL ASSIGNMENT OVERVIEW	62
30.1	OVERVIEW	62
30.2	PROJECT ROLES	62
30.2.1	THE CLIENT	62
30.2.2	THE DATA SCIENTIST	62
30.3	SUMMARY	63
31	FINAL PROJECT: CRISP-DM DATA SCIENCE METHODOLOGY	63
31.1	SELECTED TOPIC	63
31.2	BUSINESS UNDERSTANDING	63
31.2.1	PROBLEM STATEMENT	63
31.2.2	DATA QUESTION	63
31.2.3	OBJECTIVE	63
31.3	ANALYTIC APPROACH	63
31.4	DATA REQUIREMENTS	64
31.5	DATA COLLECTION PLAN	64
31.6	DATA UNDERSTANDING AND PREPARATION	64
31.6.1	DATA UNDERSTANDING	64
31.6.2	DATA PREPARATION	64
31.7	MODELING APPROACH	65

31.8	EVALUATION PLAN	65
31.9	CRISP-DM DATA REQUIREMENTS	65
32	REVIEW WHAT WE LEARNED	66
32.1	FOUNDATIONAL METHODOLOGY	66
32.2	CRISP-DM METHODOLOGY	66
32.3	BUSINESS UNDERSTANDING	66
32.4	ANALYTIC APPROACH	66
32.5	DATA REQUIREMENTS	66
32.6	DATA COLLECTION	66
32.7	DATA UNDERSTANDING	67
32.8	DATA PREPARATION	67
32.9	MODELING	67
32.10	EVALUATION	68
32.11	DEPLOYMENT	68
32.12	FEEDBACK	68

Module 1: From Problem to Approach and From Requirements to Collection

1 Data Science Methodology Overview

1.1 Learning Objectives

After completing this topic, we will be able to:

- Describe the term **methodology**.
- Relate **methodology** to **data science**.
- Explain **John Rollins'** contributions to **data science methodology**.
- Identify the **10 stages** of the standard data science methodology.
- Categorize the questions associated with the **10 stages** of the standard data science methodology.

1.2 Introduction

Data science is an influential domain that combines **statistical analysis, technological expertise, and domain knowledge** to extract valuable insights from extensive data sets.

However, despite the recent increase in computing power and easier access to data, we often do not understand the questions being asked or know how to apply the data correctly to address the problem at hand. Using a methodology helps resolve these issues.

1.3 What Is a Methodology?

A **methodology** is a system of methods used in a particular area of study.

A methodology is a guideline for the decisions researchers must make during the scientific process.

In the context of data science, **data science methodology** is a structured approach that guides data scientists in solving complex problems and making data-driven decisions.

Data science methodology also includes:

- Data collection forms
- Measurement strategies
- Comparisons of data analysis methods relative to different research goals and situations

Using a methodology provides the practical guidance needed to conduct scientific research efficiently.

There is often a temptation to bypass methodology and jump directly to solutions. However, jumping to solutions hinders our best intentions for solving problems.

1.4 Data Science Methodology

The data science methodology discussed in this course is a methodology outlined by **John Rollins**, a seasoned **IBM Senior Data Scientist**.

This course is built on his professional experience and insights into the importance of following a methodology for successful data science outcomes.

1.5 The 10 Stages of Data Science Methodology

As a general methodology, data science methodology consists of the following 10 stages:

- Business Understanding
- Analytic Approach
- Data Requirements
- Data Collection
- Data Understanding
- Data Preparation
- Modeling
- Evaluation
- Deployment
- Feedback

1.6 The Importance of Asking Questions

Asking questions is the cornerstone of success in data science.

Questions drive every stage of the data science methodology.

Data science methodology aims to answer **10 basic questions**, which align with the methodology stages.

1.6.1 Defining the Business Problem and Approach

These first two questions help define the issue and determine the appropriate approach.

- What is the problem that we are trying to solve?
- How can we use data to answer the question?

1.6.2 Organizing the Data

The next four questions help organize the data.

- What data do we need to answer the question?
- Where is the data source from, and how will we receive the data?
- Does the data we collect represent the problem to be solved?
- What additional work is required to manipulate and work with the data?

1.6.3 Validating the Solution

The final four questions help validate the approach and final design for ongoing analysis.

- When we apply data visualizations, do we see answers that address the business problem?
- Does the data model answer the initial business question, or must we adjust the data?
- Can we put the model into practice?
- Can we get constructive feedback from the data and the stakeholder to answer the business question?

1.7 Key Takeaways

- Data science methodology guides data scientists in solving complex problems using data.
- A methodology includes data collection forms, measurement strategies, and comparisons of data analysis methods relative to different research goals and situations.
- Following a structured methodology improves the effectiveness and efficiency of data science projects.
- The standard data science methodology consists of the following 10 stages:
 - Business Understanding
 - Analytic Approach
 - Data Requirements
 - Data Collection
 - Data Understanding
 - Data Preparation
 - Modeling
 - Evaluation
 - Deployment
 - Feedback
- The methodology is driven by 10 key questions that focus on:
 - Defining the business issue.
 - Determining the analytical approach.
 - Organizing and preparing the data.
 - Validating the solution and supporting continuous improvement through deployment and feedback.

2 Data Science Methodology 101: From Problem to Approach – Business Understanding

2.1 Introduction

Business Understanding is the first stage of the Data Science Methodology because it focuses on clearly understanding the problem before attempting to solve it. Taking time to clarify the business problem ensures that the correct data, analytical approach, and solution are selected. Rather than making assumptions, we should seek clarification so that our efforts address the actual business need.

2.2 Why Business Understanding Is Important

Consider a situation where we are assigned an important task with a very tight deadline. After discussing the task with our manager, we leave the meeting feeling confident that everything has been covered. However, after spending time analyzing the problem, we realize that several important questions remain unanswered.

At this point, we face two options:

- Move forward based on assumptions.
- Stop and seek clarification.

The Data Science Methodology recommends seeking clarification first. This process is known as **Business Understanding**, and it forms the foundation for every successful data science project.

2.3 Defining the Business Problem

Having a clear understanding of the problem allows us to determine:

- Which data will be required.
- Which analytical approach should be used.
- How the results will answer the core business question.

Rollins emphasizes that having a clearly defined question is essential because it ultimately determines the analytical approach required to solve the problem.

A common mistake is spending significant effort answering what people **think** is the question. Although the analytical methods used may be technically correct, they may fail to solve the actual business problem.

2.4 Understanding the Goal Before the Question

A clearly defined question begins with understanding the goal of the person asking it.

For example, suppose a business owner asks:

"How can we reduce the costs of performing an activity?"

Before answering, we need to clarify the actual goal.

Possible goals include:

- Improving the efficiency of the activity.
- Increasing the business's profitability.

Without understanding the real objective, we may solve the wrong problem.

2.5 Defining Objectives

Once the overall goal has been clarified, the next step is to identify the objectives that support that goal.

Breaking the objectives into smaller components helps us:

- Conduct structured discussions.
- Identify priorities.
- Organize the work effectively.
- Develop a plan for solving the problem.

Depending on the problem, different stakeholders should participate in these discussions to:

- Determine business requirements.
- Clarify outstanding questions.
- Ensure all perspectives are considered.

2.6 Case Study: Healthcare Budget Allocation

2.6.1 Business Problem

The case study examines the following business question:

What is the best way to allocate the limited healthcare budget to maximize its use in providing quality care?

This became a significant issue for an American healthcare insurance provider.

As public funding for hospital readmissions was decreasing, the insurance company faced the possibility of covering the additional costs. This could potentially require increasing insurance rates for customers.

Since increasing insurance rates was not considered a desirable solution, the insurance company collaborated with:

- Regional healthcare authorities.
- IBM Data Scientists.

The objective was to determine how data science could help solve the problem.

2.7 Establishing Goals and Objectives

Before collecting any data, the team focused on defining clear goals and objectives.

After discussions, the team identified **patient readmissions** as the highest-priority area for investigation.

2.8 Key Findings

With the goals and objectives established, the team discovered:

- Approximately **30%** of individuals who completed rehabilitation treatment were readmitted to a rehabilitation center within **one year**.
- Approximately **50%** were readmitted within **five years**.

Further analysis of historical records revealed that:

- Patients with **Congestive Heart Failure** were at the top of the readmission list.

2.9 Selecting the Analytical Approach

Based on the business understanding and initial findings, the team determined that a **Decision Tree model** could be applied to analyze the problem and identify why these readmissions were occurring.

2.10 IBM Data Scientists' Approach

To establish a strong business understanding before beginning the analytics work, the IBM Data Scientists proposed and delivered an **on-site workshop** to initiate the project.

The workshop helped ensure that everyone shared a common understanding of:

- The business problem.
- The project goals.
- The project objectives.
- The business requirements.

2.11 Role of the Business Sponsor

The involvement of the key business sponsor was critical throughout the project.

The sponsor:

- Set the overall direction.
- Remained actively engaged and provided guidance.
- Ensured the necessary support was available whenever required.

2.12 Business Requirements

The team identified four essential business requirements for the predictive model.

The model needed to:

- Predict readmission outcomes for patients with **Congestive Heart Failure**.
- Predict the risk of patient readmission.

- Understand the combination of events that led to the predicted outcome.
- Apply an easy-to-understand process for evaluating the readmission risk of new patients.

2.13 Key Takeaways

- Business Understanding is the first step in the Data Science Methodology.
- We should clarify the business problem before collecting or analyzing data.
- Clearly defining the business goal helps identify the correct objectives and analytical approach.
- Solving the wrong question—even with correct analytical methods—does not solve the real business problem.
- Stakeholder involvement is essential for defining goals, objectives, and business requirements.
- In the healthcare case study, Business Understanding enabled IBM Data Scientists and healthcare stakeholders to focus on patient readmissions, identify patients with Congestive Heart Failure as a high-risk group, and select a Decision Tree model as the appropriate analytical approach.

3 Data Science Methodology 101: From Problem to Approach (Analytic Approach)

3.1 Overview

Selecting the right analytic approach depends on the question being asked. The approach involves seeking clarification from the person who is asking the question so that we can select the most appropriate path or approach.

Once the problem to be addressed is defined, the appropriate analytic approach for the problem is selected in the context of the business requirements. This is the second stage of the Data Science Methodology.

Once a strong understanding of the question is established, the analytic approach can be selected. This means identifying what type of patterns will be needed to address the question most effectively.

3.2 Selecting the Appropriate Analytic Approach

The choice of analytic approach depends on the nature of the question being addressed.

- If the question is to determine the probabilities of an action, then a **predictive model** might be used.
- If the question is to show relationships, then a **descriptive approach** may be required. This approach looks at clusters of similar activities based on events and preferences.
- **Statistical analysis** applies to problems that require counts.
- If the question requires a **yes/no answer**, then a **classification approach** for predicting a response would be suitable.

3.3 Machine Learning in the Analytic Approach

Machine Learning is a field of study that gives computers the ability to learn without being explicitly programmed.

Machine Learning can be used to identify relationships and trends in data that might otherwise not be accessible or identified.

When the objective is to learn about human behaviour, an appropriate response is to use **Clustering Association approaches**.

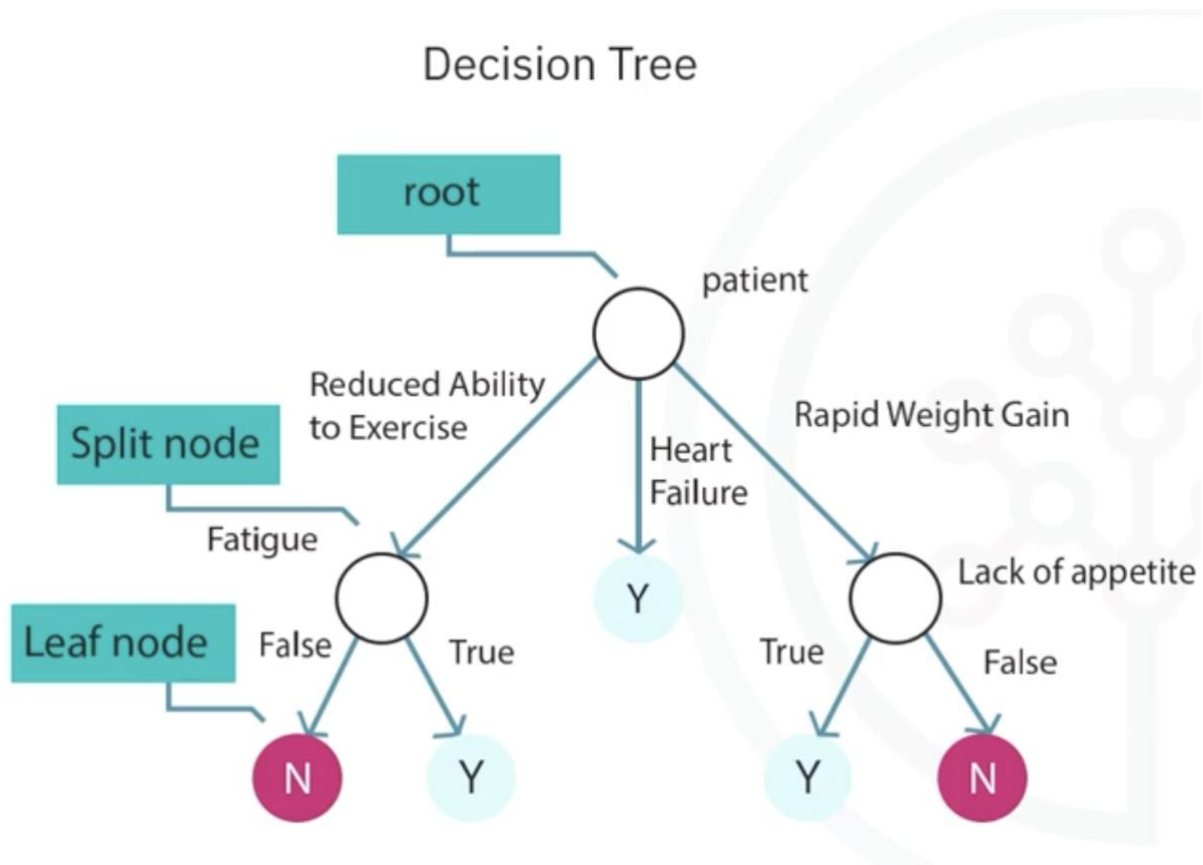
3.4 Case Study: Applying the Analytic Approach

For the case study, a **Decision Tree Classification Model** was used to identify the combination of conditions leading to each patient's outcome.

In this approach, examining the variables in each of the nodes along each path to a leaf led to a respective threshold value.

The decision tree classifier provides:

- The predicted outcome.
- The likelihood of that outcome, based on the proportion of the dominant outcome (**Yes** or **No**) in each group.



From this information, analysts can obtain the **readmission risk**, or the likelihood of a **Yes** outcome for each patient.

- If the dominant outcome is **Yes**, then the risk is simply the proportion of **Yes** patients in the leaf.
- If the dominant outcome is **No**, then the risk is **1 minus the proportion of No patients in the leaf**.

3.5 Why a Decision Tree Classification Model Was Chosen

A Decision Tree Classification Model is easy for non-data scientists to understand and apply when scoring new patients for their risk of readmission.

The approach offers several advantages:

- Clinicians can readily see which conditions are causing a patient to be classified as high-risk.
- Multiple models can be built and applied at various points during a patient's hospital stay.
- This provides a moving picture of the patient's risk and how it evolves as different treatments are applied.

For these reasons, the **Decision Tree Classification** approach was chosen for building the **Congestive Heart Failure Readmission Model**.

3.6 Key Takeaways

- Selecting the correct analytic approach depends on clearly understanding the business question.
- The analytic approach is the second stage of the Data Science Methodology.
- Different business questions require different analytical techniques, such as predictive models, descriptive analysis, statistical analysis, classification, or clustering.
- Machine Learning helps uncover relationships and trends that may not be easily identified through traditional analysis.
- Decision Tree Classification is an interpretable model that provides both predictions and their associated probabilities, making it valuable in healthcare applications such as predicting hospital readmissions.

4 Analytic Approach Based on the Question Type

When choosing an analytic approach for a problem, the type of question we are trying to answer greatly influences the methodology. Different question types require different analytic approaches, techniques, and methods to generate meaningful insights.

4.1 Descriptive Questions: "What is the Current Status?"

Approach: Descriptive Analytics

Question Example:

- "What is the current status of our sales?"

Techniques:

- Data aggregation (combining data from various sources into a unified view)
- Data mining (extracting useful information from large datasets)
- Data visualization (using visual tools to present data in an easily understandable format)

Examples:

- Summarizing sales data
- Creating dashboards
- Generating reports

4.2 Diagnostic Questions: "Why Did It Happen?"

Approach: Diagnostic Analytics

Question Example:

- "Why did our sales decline in the last quarter?"

Techniques:

- Drill-down (exploring detailed data to find underlying causes)
- Data discovery (identifying patterns and relationships in data)
- Correlation analysis (assessing the relationship between different variables)

Examples:

- Identifying root causes of sales decline
- Analyzing customer complaints
- Understanding failure points in a process

4.3 Predictive Questions: "What Is Likely to Happen?"

Approach: Predictive Analytics

Question Example:

- "What is our sales forecast for the next year?"

Techniques:

- Regression analysis (predicting outcomes based on relationships between variables)
- Time series forecasting (predicting future values based on past trends)

- Machine learning models (using algorithms to predict future outcomes based on historical data)

Examples:

- Forecasting sales
- Predicting customer churn
- Estimating future demand

4.4 Prescriptive Questions: "What Should We Do?"

Approach: Prescriptive Analytics

Question Example:

- "What should we do to increase website traffic?"

Techniques:

- Optimization models (finding the best solution from a set of alternatives)
- Simulation (modeling scenarios to predict outcomes)
- Decision analysis (evaluating and comparing different decisions)

Examples:

- Recommending inventory levels
- Optimizing marketing campaigns
- Determining pricing strategies

4.5 Classification Questions: "Which Category Does This Belong To?"

Approach: Classification (Supervised Learning)

Question Example:

- "Which category does this data point belong to?"

Techniques:

- Logistic regression (predicting the probability of a categorical outcome)
- Decision trees (splitting data into branches to classify it)
- Support Vector Machines (finding the best boundary to separate categories)
- Neural networks (using interconnected nodes to classify data)

Examples:

- Email spam detection
- Image classification
- Disease diagnosis

4.6 Summary

Understanding the different types of analytical questions and their corresponding analytic approaches helps us select the most appropriate methodology for solving business problems and extracting valuable insights from data.

Question Type	Key Question	Analytic Approach	Common Techniques	Example Applications
Descriptive	What is the current status?	Descriptive Analytics	Data aggregation, Data mining, Data visualization	Sales summaries, Dashboards, Reports
Diagnostic	Why did it happen?	Diagnostic Analytics	Drill-down, Data discovery, Correlation analysis	Root cause analysis, Customer complaints, Process analysis
Predictive	What is likely to happen?	Predictive Analytics	Regression analysis, Time series forecasting, Machine learning models	Sales forecasting, Customer churn prediction, Demand estimation
Prescriptive	What should we do?	Prescriptive Analytics	Optimization models, Simulation, Decision analysis	Inventory recommendations, Marketing optimization, Pricing strategies
Classification	Which category does this belong to?	Classification (Supervised Learning)	Logistic regression, Decision trees, Support Vector Machines, Neural networks	Spam detection, Image classification, Disease diagnosis

5 Business Understanding: Identifying Relevant Questions

5.1 Overview

The company's e-commerce business goal is to optimize its pricing strategy to maximize revenue and profitability. By leveraging data science, we aim to identify patterns in historical sales data, pricing changes, and customer behavior to make informed decisions on pricing and promotional strategies.

This topic highlights how different types of questions align with this business goal in a data science context.

5.2 Relevant Questions to the Business Goal

- How do customer purchase behaviors change during specific promotional periods?
- What is the historical website traffic data for the e-commerce site?
- What are the profit margins for different products?
- Which products have experienced the highest sales volumes in the past?

- How do product ratings and reviews influence customer purchase decisions?
- What are the customers' preferred payment methods?
- How do customer demographics influence their price sensitivity?

5.3 Questions That Are Not Relevant to the Business Goal

- How many employees work in the marketing department?
- What is the company's organizational structure?
- How much does the company spend on office supplies?

6 Analytical Approach: Identifying the Pattern to Address the Question

6.1 Overview

A transportation company aims to optimize its delivery routes and schedules to minimize costs and improve delivery efficiency. The company wants to use data science to identify the most optimal routes and delivery time windows based on historical delivery data and external factors such as traffic and weather conditions.

This topic highlights how different types of questions align with various analytical approaches.

6.2 Predictive Model

A **Predictive Model** is used when we want to forecast or estimate future outcomes based on historical data.

Examples include:

- How can we forecast the optimal number of delivery vehicles required for a specific day based on the expected order volume?
- What is the expected delivery time for each route considering historical traffic patterns and anticipated weather conditions?
- How can we anticipate the potential impact of traffic incidents or road closures on delivery times to proactively adjust routes?

6.3 Descriptive Model

A **Descriptive Model** is used to summarize historical data and provide insights into what has happened.

Examples include:

- What are the average delivery costs for different delivery routes, and how do they vary during different times of the day?
- What insights can be gathered on the average delivery times for different vehicle types, and how do these times vary based on the complexity of the delivery route?

- What are the most frequently used routes and their respective delivery time variations during peak and off-peak hours?
- What historical data highlights the busiest delivery days and time intervals during the week based on past order data?

6.4 Classification Model

A **Classification Model** is used to assign data into predefined categories or classes.

Examples include:

- What are the various time slots in which delivery schedules can be classified to balance workload and minimize delivery delays?
- How can we classify delivery routes into different categories based on the average delivery time and order volume?

6.5 Clustering (Grouping-Based Analytical Approach)

Clustering is an unsupervised analytical approach used to group similar data points together without explicitly making predictions based on historical data.

Examples include:

- How can we group delivery regions based on customer density and order frequency to optimize delivery route planning?
- How can we cluster customer locations to create distinct groups for efficient delivery route planning, without explicitly making predictions based on past data?
- How can we determine the most suitable delivery routes for perishable goods, ensuring timely deliveries without explicitly using past data to make predictions?

7 From Problem to Approach

7.1 Overview

We have completed this lesson. At this point, we have learned the key concepts of the data science methodology, including how to understand a business problem and determine the appropriate analytical approach to address it.

7.2 Business Understanding: Understanding the Question

Data science methodology begins with seeking clarification and attaining a clear business understanding.

- **Define:** Clearly defining the question is vital, as it guides the analytical approach.
- **Understand:** Understanding the goal of the person asking the question is crucial in establishing a clearly defined question.
- **Objectives:** Objectives that support the goal should be identified to prioritize and plan for problem-solving.

- **Engagement:** Engagement of different stakeholders is vital to determine requirements and clarify questions.

7.3 Analytical Approach: Determine the Appropriate Approach

The second stage of the data science methodology involves selecting the analytical approach in the context of business requirements.

- **Pattern:** Identifying what type of patterns will be needed to address the question most effectively.
- **Approach:** Different types of patterns and relationships in data require different approaches to effectively address the questions being asked.
- **Machine Learning:** Machine learning enables computers to learn without explicit programming and identify patterns in data that might otherwise not be accessible.

7.4 Key Takeaway

During the **Analytical Approach** stage, we can choose from the following analytical approaches based on the business problem:

- Descriptive Analytics
- Diagnostic Analytics
- Predictive Analytics
- Prescriptive Analytics

We also determine whether to use machine learning techniques such as:

- Clustering
- Association Analysis

8 Glossary: From Problem to Approach

8.1 Overview

This alphabetized glossary contains many of the terms we will encounter in this lesson. These terms are important to recognize when working in the industry.

8.2 Glossary

Term	Definition
Analytic Approach	The process of selecting the appropriate method or path to address a specific data science question or problem.
Analytics	The systematic analysis of data using statistical, mathematical, and computational techniques to uncover insights, patterns, and trends.
Business Understanding	The initial phase of the data science methodology that involves seeking clarification and understanding the goals, objectives, and requirements of a given task or problem.
Clustering Association	An approach used to learn about human behavior and identify patterns and associations in data.

Cohort	A group of individuals who share a common characteristic or experience and are studied or analyzed as a unit.
Cohort Study	An observational study where a group of individuals with a specific characteristic or exposure is followed over time to determine outcomes or relationships.
Congestive Heart Failure (CHF)	A chronic condition in which the heart cannot pump enough blood to meet the body's needs, resulting in fluid buildup and symptoms such as shortness of breath and fatigue.
CRISP-DM	Cross-Industry Standard Process for Data Mining , a widely used methodology encompassing six phases: Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment.
Data Analysis	The process of inspecting, cleaning, transforming, and modeling data to discover useful information, draw conclusions, and support decision-making.
Data Cleansing	The process of identifying and correcting or removing errors, inconsistencies, or inaccuracies in a dataset to improve its quality and reliability.
Data Science	An interdisciplinary field that combines scientific methods, processes, algorithms, and systems to extract knowledge and insights from data.
Data Science Methodology	A structured approach to solving business problems using data analysis and data-driven insights.
Data Scientist	A professional who uses scientific methods, algorithms, and tools to analyze data, extract insights, and develop solutions to complex problems.
Data Scientists	Professionals with data science and analytics expertise who apply their skills to solve business problems.
Data-Driven Insights	Insights derived from analyzing and interpreting data to inform decision-making.
Decision Tree	A supervised machine learning algorithm that uses a tree-like structure of decisions to make predictions or classifications.
Decision Tree Classification Model	A model that classifies data based on conditions and thresholds, providing predicted outcomes and probabilities.
Decision Tree Classifier	A classification model that determines outcomes based on specific conditions and thresholds.
Decision-Tree Model	A model used to review scenarios and identify relationships in data.
Descriptive Approach	An approach used to show relationships and identify clusters based on events and preferences.
Descriptive Modeling	A modeling technique focused on describing and summarizing data without making predictions.
Domain Knowledge	Expertise and understanding of a specific subject area, including its concepts and relevant data.
Goals and Objectives	The desired outcomes and specific objectives that support the overall goal of a task or problem.

Iteration	A single cycle or repetition of a process involving refinement based on feedback or new information.
Iterative Process	A process that involves repeating steps to refine and improve a solution or analysis.
Leaf	The final nodes of a decision tree where data is categorized into specific outcomes.
Machine Learning	A field that enables computers to learn from data without explicit programming, identifying hidden patterns.
Mean	The average value of a set of numbers.
Median	The middle value in a sorted dataset.
Model (Conceptual Model)	A simplified representation of a real-world system used to analyze or predict behavior.
Model Building	The process of developing predictive models to gain insights and support decisions.
Pairwise Comparison (Correlation)	A statistical technique measuring the relationship between two variables.
Pattern	A recurring arrangement or sequence in data that provides insights.
Predictive Model	A model used to determine probabilities of outcomes based on historical data.
Predictors	Variables used in a model to predict an outcome.
Prioritization	Organizing objectives based on importance and impact.
Problem Solving	The process of addressing challenges to achieve desired outcomes.
Stakeholders	Individuals or groups with an interest in the outcome of a data science model.
Standard Deviation	A measure of variability or dispersion in a dataset.
Statistical Analysis	The application of statistical methods to analyze data and solve problems.
Statistics	The study of data collection, analysis, interpretation, and presentation.
Structured Data (Data Model)	Data organized according to a predefined schema, typically stored in databases.
Text Analysis Data Mining	The process of extracting information from unstructured text using techniques like Natural Language Processing (NLP) (a branch of AI that enables computers to understand and process human language).
Threshold Value	A value used to split data into categories in a decision tree.

9 Data Science Methodology: Data Requirements

9.1 Introduction

Data requirements are a critical stage in the Data Science Methodology. Before collecting or preparing data, we must determine exactly what data is needed to solve the problem. Just as preparing a spaghetti dinner requires the right ingredients, solving a data science problem requires the right data. If the required data is missing or inappropriate, the success of the project may be compromised.

If the problem to be solved is considered the recipe, then data is the ingredient. Therefore, we must identify:

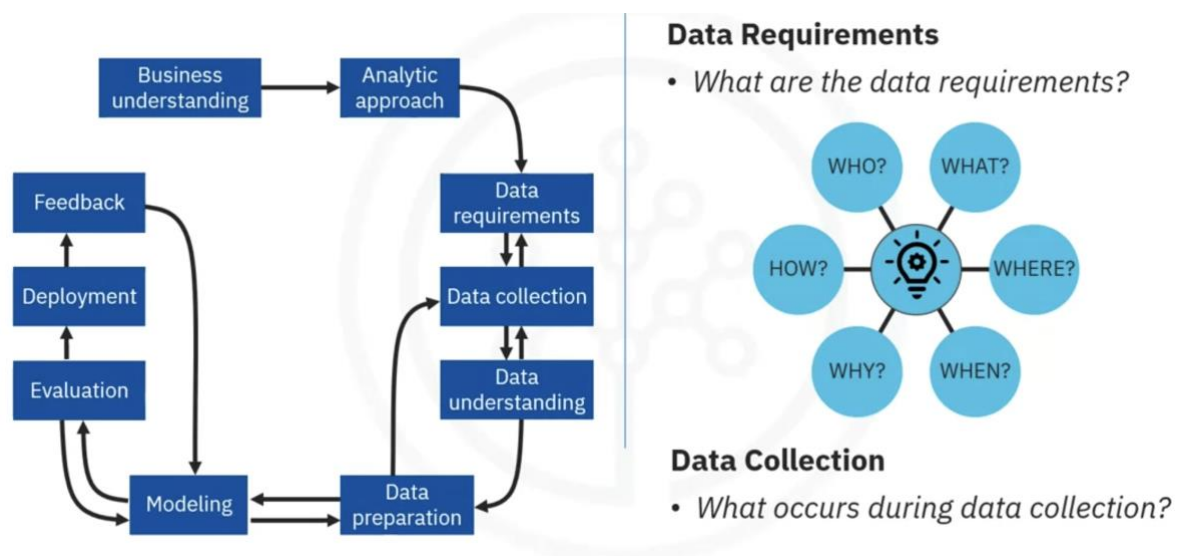
- Which data is required.
- How to source or collect the data.
- How to understand and work with the data.
- How to prepare the data to achieve the desired outcome.

Building on our understanding of the problem and the selected analytical approach, we are ready to define the data requirements.

9.2 Defining Data Requirements

Before beginning the **Data Collection** and **Data Preparation** stages, it is essential to define the data requirements for the selected analytical approach. For a **Decision Tree Classification** model, this includes identifying:

- The necessary data content.
- The required data formats.
- The appropriate data sources for the initial data collection.



9.3 Case Study: Applying Data Requirements

In the case study, the first task was to define the data requirements for the **Decision Tree Classification** approach that had been selected.

This included selecting a suitable patient cohort from the health insurance provider's member base.

9.3.1 Patient Cohort Selection Criteria

To compile complete clinical histories, three criteria were identified for including patients in the cohort:

- The patient needed to be admitted as an **inpatient** within the provider's service area so they would have access to the necessary information.
- The patient must have had a **primary diagnosis of congestive heart failure** during one full year.
- The patient must have had **continuous enrollment for at least six months** prior to the primary admission for congestive heart failure so that a complete medical history could be compiled.

Patients with congestive heart failure who had also been diagnosed with other significant medical conditions were excluded from the cohort because those conditions would cause **higher-than-average re-admission rates**, which could skew (distort or bias) the results.

9.4 Data Content and Format Requirements

After selecting the patient cohort, the required content, format, and representation of the data for Decision Tree Classification were defined.

This modeling technique requires:

- One record per patient.
- Columns representing the variables used in the model.

To model the readmission outcome, data was required to cover all aspects of each patient's clinical history.

The required content included:

- Admissions.
- Primary diagnoses.
- Secondary diagnoses.
- Tertiary diagnoses.
- Procedures.
- Prescriptions.
- Other healthcare services provided during hospitalization.
- Services provided throughout patient-doctor visits.

A single patient could have **thousands of records**, representing all related clinical attributes.

9.5 Preparing Data for Modeling

Since the Decision Tree model requires **one record per patient**, the transactional records needed to be transformed.

The data scientists:

- Rolled up (aggregated) the transactional records to the patient level.
- Created a number of new variables to represent the summarized information.

This transformation is performed during the **Data Preparation** stage.

This example highlights the importance of thinking ahead and anticipating the requirements of subsequent stages in the Data Science Methodology when defining data requirements.

10 Data Collection

Data collection is the stage that follows the **Data Requirements** phase in the Data Science Methodology. After the initial data collection is performed, we assess whether we have all the data required to solve the problem.

Similar to shopping for ingredients to prepare a meal, some data may be unavailable, difficult to obtain, or more expensive to acquire than initially expected. Therefore, during this stage, the data requirements may be revised, and decisions are made about whether more data is needed, less data is sufficient, or alternative data sources should be considered.

Once the required data has been collected, we gain a clear understanding of the data that will be used throughout the project.

10.1 Assessing the Collected Data

After collecting the data, we evaluate its overall quality and suitability for analysis.

The following techniques are commonly applied:

- **Descriptive statistics** to summarize the characteristics of the dataset.
- **Data visualization** to assess the content, quality, and obtain initial insights from the data.
- Identification of **gaps or missing data**, followed by planning to either fill those gaps or make suitable substitutions.

At this stage, the collected data is ready for preparation and further analysis, similar to having all the ingredients placed on the cutting board before cooking begins.

10.2 Case Study: Applying Data Collection

Data collection requires knowing the source of the required data elements and where they can be obtained.

In the context of the case study, the required data included:

- Demographic information of patients.
- Clinical information of patients.
- Patient coverage information.
- Provider information.
- Claims records.
- Pharmaceutical information.
- Other information related to all diagnoses of patients with **congestive heart failure**.

For this case study, certain **drug information** was also required. However, that data source had not yet been integrated with the other available data sources.

10.3 Handling Unavailable Data

An important principle of the data collection stage is that it is acceptable to defer decisions regarding unavailable data and attempt to acquire it later in the project.

For example:

- We may first build an initial predictive model using the available data.
- After obtaining intermediate modeling results, we can evaluate whether the missing drug information is important for improving the model.
- If the results indicate that the additional data would significantly improve model performance, then the effort to acquire it is justified.

In this particular case study, the team was able to build a reasonably good predictive model without including the unavailable drug information.

10.4 Data Integration

Database Administrators (DBAs) and programmers often collaborate to:

- Extract data from multiple data sources.
- Merge data from different systems.
- Remove redundant data.
- Prepare the integrated dataset for the next stage of the Data Science Methodology.

This integrated dataset is then ready for the **Data Understanding** phase.

10.5 Improving the Data Collection Process

At this stage, if necessary, data scientists and analytics team members can discuss methods to improve data management.

Possible improvements include:

- Automating database processes.
- Streamlining data extraction procedures.
- Making future data collection easier and faster.

11 From Requirements to Collection

11.1 Overview

We have completed this lesson. At this point, we understand the importance of identifying the right data requirements and ensuring appropriate data collection to support the analytical approach.

The **Data Requirements** stage includes identifying the correct and necessary:

- Data content

- Data formats
- Data sources

These requirements are determined based on the specific analytical approach.

During the **Data Collection** stage, expert data scientists meticulously revise data requirements and make critical decisions regarding the **quantity** and **quality** of data.

11.2 Data Requirements and Collection: Ensuring the Right Ingredients

This stage of the data science methodology emphasizes identifying, sourcing, understanding, and preparing the necessary data for analysis.

11.2.1 Data Requirements

The initial data collection is assessed to determine whether it fulfills the defined requirements.

Based on the availability, quality, and content of the collected data, the requirements are revised as needed.

11.2.2 Data Collection

The Data Collection stage involves:

- Identifying the source of data to obtain the required data elements.
- Applying appropriate data acquisition strategies.
- Making decisions regarding unavailable data.

11.2.3 Data Integration

Database Administrators (DBAs) and programmers perform the following tasks:

- Extract data from diverse sources.
- Merge data from multiple sources.
- Eliminate redundant data.
- Prepare the integrated data for the Data Understanding stage.

Database processes may also be automated to improve efficiency and speed.

11.2.4 Data Understanding

Data scientists use descriptive statistics and visualization techniques to assess the collected data.

The main objectives are to:

- Assess the data content and quality.
- Gain initial insights from the collected data.
- Identify data gaps.

- Plan how missing data will be filled or substituted.
- Prepare the data for the subsequent stages of analysis.

11.3 Key Takeaway

Data scientists apply **descriptive statistics** and **visualization techniques** to:

- Assess the content and quality of collected data.
- Obtain initial insights from the data.
- Identify data gaps.
- Determine whether additional data is required.
- Decide whether existing data should be substituted when necessary.

12 Glossary: From Requirements to Collection

12.1 Overview

This alphabetized glossary contains many of the terms we will encounter in this topic.

12.2 Glossary

Term	Definition
Analytics Team	A group of professionals, including data scientists and analysts, responsible for performing data analysis and modeling.
Data Collection	The process of gathering data from various sources, including demographic, clinical, coverage, and pharmaceutical information.
Data Integration	The merging of data from multiple sources to remove redundancy and prepare it for further analysis.
Data Preparation	The process of organizing and formatting data to meet the requirements of the modeling technique.
Data Requirements	The identification and definition of the necessary data elements, formats, and sources required for analysis.
Data Understanding	A stage where data scientists discuss various ways to manage data effectively, including automating certain processes in the database.
DBAs (Database Administrators)	Professionals responsible for managing and extracting data from databases.
Decision Tree Classification	A modeling technique that uses a tree-like structure to classify data based on specific conditions and variables.
Demographic Information	Information about patient characteristics, such as age, gender, and location.
Descriptive Statistics	Techniques used to analyze and summarize data, providing initial insights and identifying gaps in data.
Intermediate Results	Partial results obtained from predictive modeling that can influence decisions on acquiring additional data.
Patient Cohort	A group of patients with specific criteria selected for analysis in a study or model.
Predictive Modeling	The process of building models to predict future outcomes based on historical data.

Training Set	A subset of data used to train or fit a machine learning model. It consists of input data and corresponding labeled outputs.
Unavailable Data	Data elements that are not currently accessible or integrated into available data sources.
Univariate Modeling	Analysis focused on a single variable or feature at a time.
Unstructured Data	Data that does not have a predefined structure or format (e.g., text, images, audio, or video) and requires specialized techniques for analysis.
Visualization	The process of representing data visually to gain insights into its content and quality.

Module 2: From Understanding to Preparation and From Modeling to Evaluation

13 Data Science Methodology 101: Data Understanding

13.1 Introduction

Data understanding encompasses all activities related to constructing the dataset. Essentially, the data understanding stage of the Data Science Methodology answers the following question:

Is the data that we collected representative of the problem to be solved?

13.2 Data Understanding in the Congestive Heart Failure Case Study

To understand the data related to **congestive heart failure admissions**, descriptive statistics were run against the data columns that would become variables in the model.

The analysis included the following steps:

- Hurst (as mentioned in the source), univariates, and statistics on each variable.
- Statistical measures such as:
 - Mean
 - Median
 - Minimum
 - Maximum
 - Standard deviation

These statistics helped us understand the characteristics of each variable before building the model.

13.3 Pairwise Correlation Analysis

Pairwise correlations were used to determine:

- How closely certain variables were related.

- Which variables were highly correlated with one another.

Very highly correlated variables are essentially redundant(not necessary), making only one of them relevant for modeling.

13.4 Using Histograms to Understand Data Distribution

Histograms of the variables were examined to understand their distributions.

Histograms help us understand:

- How the values of a variable are distributed.
- What type of data preparation may be required to make the variable more useful in a model.

For example, if a categorical variable has too many distinct values to be informative in a model, the histogram helps determine how to consolidate those values.

13.5 Assessing Data Quality

The univariates, statistics, and histograms were also used to assess data quality.

Based on the information provided, certain values could be:

- Re-coded.
- Dropped if necessary, such as when a variable contains missing values.

An important question to ask is:

Does "missing" mean anything?

The meaning of missing values depends on the context. For example:

- Sometimes a missing value may mean "**No**".
- Sometimes it may represent **0 (zero)**.
- In other situations, it simply means "**We don't know.**"

A variable may also contain invalid or misleading values.

For example:

- A numeric variable called **Age** may contain values from **0 to 100**, but also contain **999**.
- In this case, **999** actually means "**missing**" rather than a valid age.
- Unless corrected during data preparation, **999** would incorrectly be treated as a valid value by the model.

13.6 Refining the Definition of Congestive Heart Failure Admission

Initially, the meaning of **congestive heart failure admission** was defined based only on a **primary diagnosis** of congestive heart failure.

However, working through the data understanding stage revealed that this initial definition was not capturing all expected congestive heart failure admissions, based on clinical experience.

As a result, we returned to the **Data Collection** stage and:

- Added secondary diagnoses.
- Added tertiary diagnoses.
- Built a more comprehensive definition of congestive heart failure admission.

13.7 The Iterative Nature of Data Science Methodology

This example demonstrates that the Data Science Methodology is an interactive and iterative process.

The more we work with the problem and the data:

- The more we learn.
- The more refinements we can make.
- The better the resulting model becomes.

These continuous improvements ultimately lead to a better solution to the problem.

14 Data Science Methodology 101: Data Preparation – Concepts

14.1 Introduction

Data preparation is similar to washing freshly picked vegetables, where unwanted elements such as dirt or imperfections are removed before use. Likewise, in data science, raw data must be cleaned and transformed before it can be effectively analyzed.

Together with **Data Collection** and **Data Understanding**, **Data Preparation** is the most time-consuming phase of a data science project.

14.2 Importance of Data Preparation

- Data preparation typically takes **70% to 90%** of the overall project time.
- Automating parts of the data collection and preparation process in the database can reduce this effort to as little as **50%**.
- The time saved through automation allows data scientists to spend more time creating and improving machine learning models.

14.3 Transforming Data

To continue with the cooking metaphor, chopping onions into finer pieces allows their flavors to spread through a sauce more easily than dropping a whole onion into the pot.

Similarly, transforming data during the data preparation phase puts the data into a state where it is easier to work with and analyze.

The data preparation stage answers the following question:

What are the ways in which data is prepared?

14.4 Preparing Data for Analysis

To work effectively with data, we must prepare it in a way that:

- Addresses missing values.
- Handles invalid values.
- Removes duplicate records.
- Ensures that all data is properly formatted.

These steps improve data quality and make it suitable for further analysis and modeling.

14.5 Feature Engineering

Feature engineering is an important part of data preparation.

It is the process of using **domain knowledge** (knowledge about the specific problem or field) to create features that help machine learning algorithms perform more effectively.

A feature is a characteristic that can help to solve a problem.

Features within the data are important to predictive models and directly influence the results we want to achieve.

Feature engineering is especially critical when machine learning tools are being applied to analyze data.

14.6 Text Analysis During Data Preparation

When working with text data, text analysis steps for coding the data are required so that we can manipulate the data effectively.

The data scientist needs to know what they are looking for within the dataset to address the question being investigated.

Text analysis is critical because it helps ensure that:

- Proper groupings are established.
- The programming does not overlook important information hidden within the text.

14.7 Importance of Completing Data Preparation Carefully

The data preparation phase sets the stage for the next steps in addressing the question.

Although this phase may take considerable time, completing it correctly ensures that the results support the overall project.

If this phase is skipped or performed poorly, the outcome will not be up to the required standard and may require returning to the beginning of the project to correct the issues.

It is vital to take sufficient time during this phase and use available tools to automate common tasks, helping accelerate the data preparation process while maintaining data quality.

14.8 Key Takeaway

Pay close attention to detail during data preparation.

After all, it takes just **one bad ingredient** to ruin a fine meal.

15 Data Preparation – Case Study

Data preparation is similar to washing freshly picked vegetables, where unwanted elements such as dirt or imperfections are removed before use. In this topic, we apply the key concepts of **Data Preparation** through a healthcare case study focused on **congestive heart failure (CHF)**.

15.1 Defining Congestive Heart Failure

An important first step in the data preparation stage was to define **congestive heart failure (CHF)** precisely. Although this initially seemed straightforward, defining it accurately was not simple.

The following activities were required:

- Identify the appropriate **diagnosis-related group (DRG) codes** (codes used to classify hospital cases for diagnosis and treatment), since congestive heart failure implies certain kinds of fluid buildup.
- Consider that **congestive heart failure is only one type of heart failure**.
- Obtain **clinical guidance** from healthcare experts to ensure that the correct diagnosis codes for congestive heart failure were selected.

15.2 Defining Readmission Criteria

The next step involved defining the **readmission criteria** for congestive heart failure.

To accomplish this:

- The timing of events was evaluated to determine whether a particular congestive heart failure admission was:
 - An **initial event**, called an **index admission** (the patient's first qualifying hospitalization).
 - A **congestive heart failure-related readmission**.
- Based on clinical expertise, a **30-day period** was established as the readmission window following discharge from the initial hospitalization.

15.3 Aggregating Transactional Data

The available data was initially stored in a **transactional format**, meaning each patient had multiple records rather than a single consolidated record.

These transactional records included:

- Professional provider facility claims submitted for:
 - Physicians
 - Laboratories
 - Hospitals
 - Clinical services
- Diagnoses
- Procedures
- Prescriptions
- Other information related to both **inpatients** and **outpatients**

Depending on a patient's clinical history, a single patient could have **hundreds or even thousands of records**.

To prepare the data for modeling:

- All transactional records were **aggregated to the patient level**, producing **one record per patient**.
- This structure was required for the **decision-tree classification method** used during predictive modeling.

15.4 Creating Patient-Level Features

During the aggregation process, many new columns were created to summarize the transactional information for each patient.

Examples of these newly created variables included:

- Frequency of visits to doctors, clinics, and hospitals.
- Most recent visits.
- Diagnoses.
- Procedures.
- Prescriptions.

- Other relevant clinical information.

These aggregated features became the input variables for predictive modeling.

15.5 Considering Co-Morbidities

The data preparation process also considered **co-morbidities** (other diseases or medical conditions that occur alongside the primary disease).

Examples included:

- Diabetes
- Hypertension
- Many other chronic diseases and conditions

These conditions were included because they could significantly impact the **risk of congestive heart failure readmission**.

15.6 Literature Review and Data Collection Refinement

During discussions on data preparation, a **literature review** (review of published medical research) on congestive heart failure was conducted to determine whether any important data elements had been overlooked.

The review identified additional indicators that needed to be included.

As a result:

- The process **looped back to the Data Collection stage**.
- Additional indicators for conditions and procedures were added before continuing with data preparation.

15.7 Merging Patient Information

After aggregating the transactional data at the patient level, it was merged with other patient information, including demographic data such as:

- Age
- Gender
- Type of insurance
- Other demographic attributes

This resulted in **one comprehensive table**, where:

- Each patient was represented by a **single record**.
- Each column represented an attribute describing the patient's demographic and clinical history.

These columns served as the variables for predictive modeling.

15.8 Target Variable

The **dependent variable (target variable)** was:

Congestive heart failure readmission within 30 days following discharge from a hospitalization for congestive heart failure.

Possible outcomes:

- Yes
- No

15.9 Final Prepared Dataset

The data preparation stage resulted in:

- **2,343 patients** meeting all of the criteria for the case study.

The prepared cohort was then divided into:

- **Training set** – Used to build the predictive model.
- **Testing set** – Used to validate the predictive model.

16 Lesson Summary: From Understanding to Preparation

16.1 Overview

We have completed this lesson. At this point in the course, we know:

- The **Data Understanding** stage encompasses all activities related to constructing the dataset and answers the question of whether the collected data represents the problem to be solved.
- During the **Data Understanding** stage, data scientists might use **descriptive statistics**, **predictive statistics**, or both.
- Data scientists commonly apply **univariate statistics** (statistical analysis of a single variable), along with measures such as:
 - Mean
 - Median
 - Minimum
 - Maximum
 - Standard Deviation
 - Pairwise Correlation
 - Histograms
- Data scientists also use **univariate statistics** and **histograms** to assess data quality.

16.2 Data Understanding and Preparation

16.2.1 Data Understanding and Iterative Assessment

Data understanding and iterative assessment play a significant role in achieving successful analytical outcomes through effective data assessment and preparation techniques.

16.2.2 Importance of Data Understanding

Data understanding assesses the quality and significance of data components by using:

- Visualizations
- Descriptive statistics

16.2.3 Assessment and Iteration

Iterations (repeated cycles of refining and improving) help refine:

- The problem definition.
- The data collection methods.

16.3 Data Preparation

Data preparation involves transforming data by addressing issues such as:

- Missing values
- Invalid values
- Incorrect formatting
- Duplicate records
- Feature engineering

These activities set the stage for effective model building and analysis.

16.4 Efficiency and Quality

Data preparation also focuses on improving efficiency and ensuring high-quality data by:

- Automating data preparation processes.
- Prioritizing modeling and problem-solving activities.
- Applying feature engineering to improve model performance.
- Performing text analysis to enhance data quality and performance.

The overall focus is on ensuring precise and reliable analytical outcomes.

16.5 Key Takeaways

- During the **Data Preparation** stage, data scientists must:
 - Address missing or invalid values.
 - Remove duplicate records.

- Validate that the data is properly formatted.
- **Feature engineering**, also part of the Data Preparation stage, uses domain knowledge of the data to create features that enable machine learning algorithms to perform more effectively.
- **Text analysis** during the Data Preparation stage is critical for validating that the proper groupings are established and that the programming is not overlooking hidden data.

17 Module 2 Lesson 1 Glossary: From Understanding to Preparation

17.1 Overview

This alphabetically organized glossary contains many of the important terms covered in this lesson. These terms are essential for understanding data science concepts, working in the industry, and participating in user groups.

17.2 Glossary

Term	Definition
Automation	Using tools and techniques to streamline data collection and preparation processes.
Data Collection	The phase of gathering and assembling data from various sources.
Data Compilation	The process of organizing and structuring data to create a comprehensive data set.
Data Formatting	The process of standardizing the data to ensure uniformity and ease of analysis.
Data Manipulation	The process of transforming data into a usable format.
Data Preparation	The phase where data is cleaned, transformed, and formatted for further analysis, including feature engineering and text analysis.
Data Preparation	The stage where data is transformed and organized to facilitate effective analysis and modeling.
Data Quality Assessment	The assessment of data integrity and completeness, addressing missing, invalid, or misleading values.
Data Quality Assessment	The evaluation of data integrity, accuracy, and completeness.
Data Set	A collection of data used for analysis and modeling.
Data Understanding	The stage in the data science methodology focused on exploring and analyzing the collected data to ensure that the data is representative of the problem to be solved.

Descriptive Statistics	Summary statistics that data scientists use to describe and understand the distribution of variables, such as mean, median, minimum, maximum, and standard deviation .
Feature	A characteristic or attribute within the data that helps in solving the problem.
Feature Engineering	The process of creating new features or variables based on domain knowledge (expert knowledge of the application area) to improve machine learning algorithms' performance.
Feature Extraction	Identifying and selecting relevant features or attributes from the data set.
Interactive Processes	Iterative (repeated refinement based on previous results) and continuous refinement of the methodology based on insights and feedback from data analysis.
Missing Values	Values that are absent or unknown in the data set, requiring careful handling during data preparation.
Model Calibration	Adjusting model parameters to improve accuracy and alignment with the initial design.
Pairwise Correlations	An analysis to determine the relationships and correlations between different variables.
Text Analysis	Steps to analyze and manipulate textual data, extracting meaningful information and patterns.
Text Analysis Groupings	Creating meaningful groupings and categories from textual data for analysis.
Visualization Techniques	Methods and tools that data scientists use to create visual representations or graphics that enhance the accessibility and understanding of data patterns, relationships, and insights.

18 Data Science Methodology: Modeling – Concepts

18.1 Introduction

Modeling is the stage in the Data Science Methodology where we have the opportunity to evaluate whether our solution is effective or whether it requires further refinement. Similar to tasting a dish while cooking, this stage helps us determine if the model is accurate or if it needs additional adjustments.

This topic focuses on answering two key questions:

- What is the purpose of data modeling?

- What are some characteristics of the data modeling process?

18.2 Purpose of Data Modeling

Data Modeling focuses on developing models that are either:

- **Descriptive models**
- **Predictive models**

18.2.1 Descriptive Models

A descriptive model examines relationships and patterns within data.

Example:

- If a person did this, then they are likely to prefer that.

Descriptive models help us understand existing patterns and behaviors in the data.

18.2.2 Predictive Models

A predictive model attempts to produce outcomes such as:

- Yes/No
- Stop/Go

These models are designed to predict future outcomes based on historical data.

The models are developed based on the selected analytical approach, which may be:

- Statistically driven
- Machine learning driven

18.3 Training Set for Predictive Modeling

For predictive modeling, we use a **training set**.

A training set is a set of historical data in which the outcomes are already known.

The training set acts as a gauge to determine whether the model needs to be calibrated.

During this stage, we experiment with different algorithms to ensure that the variables being used are actually required for building an effective model.

18.4 Importance of Data Preparation and Modeling

The success of data compilation, preparation, and modeling depends on:

- Understanding the problem at hand.
- Selecting the appropriate analytical approach.

The data supports answering the question and, like the quality of ingredients in cooking, sets the stage for the final outcome.

Throughout this process, constant refinement, adjustments, and tweaking are necessary within each step to ensure the outcome is reliable and robust.

18.5 John Rollins' Descriptive Data Science Methodology

In **John Rollins'** descriptive Data Science Methodology, the framework is designed to accomplish three objectives:

- Understand the question at hand.
- Select an analytic approach or method to solve the problem.
- Obtain, understand, prepare, and model the data.

The ultimate goal is to move us to a point where a data model can be built to answer the original question.

18.6 Moving Toward Model Evaluation

Once the model has been developed, the next stage involves:

- Model evaluation
- Deployment
- Feedback loops

These stages help ensure that the answer produced by the model is both accurate and relevant.

18.7 Importance of Relevance in Data Science

Relevance is critical to the field of Data Science because it is still a relatively new field of study, and we are interested in exploring the possibilities it has to offer.

The more people who benefit from the outcomes of this practice, the further the field will continue to develop.

18.8 Summary

In this topic, we reviewed the key concepts related to the **Modeling** stage of the Data Science Methodology.

The main points covered include:

- The purpose of data modeling.
- The difference between descriptive and predictive models.

- The role of training data in predictive modeling.
- The importance of selecting the appropriate analytical approach.
- The need for continuous refinement throughout the modeling process.
- John Rollins' three-part descriptive Data Science Methodology framework.
- The transition from modeling to evaluation, deployment, and feedback.
- The importance of relevance and continuous improvement in advancing the field of Data Science.

19 Modeling – Case Study (Data Science Methodology)

19.1 Introduction

Modeling is the stage in the Data Science Methodology where we have the opportunity to evaluate whether the model performs well or requires further improvement. One of the key activities during this stage is **parameter tuning** (adjusting model settings to improve performance).

This topic applies a case study to the modeling stage using a **Decision Tree Classification** model to predict **congestive heart failure (CHF) readmission**.

19.2 Building the Initial Model

With a prepared training dataset, the first Decision Tree Classification model for **congestive heart failure readmission** can be built.

The objective is to identify patients who are at **high risk of readmission**, where the outcome of interest is:

- **Congestive heart failure readmission = "Yes"**

19.2.1 Results of the First Model

- **Overall Accuracy: 85%**
- **Accuracy for predicting "Yes" (Actual Readmissions): 45%**

Although the overall accuracy of **85%** appears to be good, the model correctly identifies only **45%** of the actual readmission cases. This means the model is not sufficiently accurate for predicting patients who will actually be readmitted.

The next question becomes:

- **How can we improve the model's accuracy in predicting the "Yes" outcome?**

19.3 Improving the Model Through Parameter Tuning

For a **Decision Tree Classification** model, one of the most effective parameters to adjust is the:

- **Relative cost of misclassified "Yes" and "No" outcomes**

The idea behind this adjustment is that the two types of prediction errors do not have the same consequences.

19.3.1 Type I Error (False Positive)

When a patient who **would not be readmitted** is incorrectly classified as a readmission case, action may be taken unnecessarily to reduce that patient's risk.

The cost of this error is:

- Unnecessary intervention
- Wasted healthcare resources

Statisticians refer to this as:

- **Type I Error (False Positive)**

19.3.2 Type II Error (False Negative)

When a patient who **would actually be readmitted** is incorrectly classified as **not** being at risk, no preventive action is taken.

The cost of this error includes:

- Hospital readmission
- Additional healthcare costs
- Trauma to the patient

Statisticians refer to this as:

- **Type II Error (False Negative)**

Since these two types of errors have different consequences, it is reasonable to assign different costs to misclassifying "Yes" and "No" outcomes.

19.4 Default Relative Cost

By default, the Decision Tree algorithm assigns an equal cost:

- **1 : 1** (Yes : No)

However, the algorithm allows us to assign a higher weight to the "Yes" outcome if correctly identifying readmission cases is considered more important.

19.5 Second Model

For the second model, the relative cost was changed to:

- **9 : 1**

This is a very high ratio intended to better understand the model's behavior.

19.5.1 Results

- **Accuracy for "Yes": 97%**
- **Overall Accuracy: 49%**

Although the model correctly classified **97%** of the readmission cases, the accuracy for predicting **"No"** became very poor, reducing the overall accuracy to only **49%**.

This was clearly **not a good model**.

The main issue was the large number of **False Positives**, which resulted in recommending unnecessary and costly interventions for patients who would not have been readmitted.

Therefore, we need to search for a better balance between predicting **"Yes"** and **"No"** outcomes.

19.6 Third Model

For the third model, the relative cost was adjusted to a more reasonable value:

- **4 : 1**

19.6.1 Results

- **Sensitivity (Accuracy for "Yes"): 68%**
- **Specificity (Accuracy for "No"): 85%**
- **Overall Accuracy: 81%**

This provided the best balance achievable with the relatively small training dataset.

19.6.2 Performance Summary

Metric	Result
Relative Cost	4 : 1
Sensitivity (Yes Accuracy)	68%
Specificity (No Accuracy)	85%
Overall Accuracy	81%

This demonstrates that adjusting the **relative cost of misclassified "Yes" and "No" outcomes** can significantly improve the balance between identifying actual readmissions and avoiding unnecessary interventions.

Model	Relative Cost Y:N	Overall Accuracy (% correct Y & N)	Sensitivity (Y accuracy)	Specificity (N accuracy)
→ 1	1:1	85%	45%	97%
→ 2	9:1	49%	97%	35%
→ 3	4:1	81%	68%	85%

19.7 Iterative Nature of Modeling

Modeling is an iterative process. Improving a model often requires returning to the **Data Preparation** stage to redefine variables so they better represent the underlying information.

By refining the input variables and continuing parameter tuning, we can further improve model performance.

19.8 Key Takeaways

- Modeling involves building and improving predictive models through parameter tuning.
- The case study uses a **Decision Tree Classification** model to predict **congestive heart failure readmission**.
- The initial model achieved **85% overall accuracy**, but only **45% accuracy** in predicting actual readmissions.
- Decision Trees can improve performance by adjusting the **relative cost of misclassified "Yes" and "No" outcomes**.
- Type I Error (False Positive):** A non-readmitted patient is predicted as readmitted, resulting in unnecessary intervention.
- Type II Error (False Negative):** A readmitted patient is predicted as not at risk, resulting in readmission, additional costs, and patient trauma.
- The default relative cost is **1 : 1**.
- Setting the relative cost to **9 : 1** increased **Yes accuracy to 97%**, but reduced overall accuracy to **49%**, producing too many false positives.
- Setting the relative cost to **4 : 1** produced the best balance:
 - Sensitivity: 68%**
 - Specificity: 85%**
 - Overall Accuracy: 81%**
- Modeling is an iterative process that often requires revisiting the **Data Preparation** stage to improve variables and enhance model performance.

20 Model Evaluation

Model evaluation goes hand-in-hand with model building. Therefore, the **modeling** and **evaluation** stages are performed iteratively.

Model evaluation is carried out during model development and before the model is deployed. It allows us to assess the quality of the model and determine whether it satisfies the original business or analytical objective.

The primary question addressed during model evaluation is:

- Does the model really answer the initial question?
- Or does the model need to be adjusted?

20.1 Phases of Model Evaluation

Model evaluation can have two main phases:

- **Diagnostic Measures**
- **Statistical Significance Testing**

20.1.1 Diagnostic Measures

The first phase is the **diagnostic measures** phase, which is used to ensure that the model is working as intended.

The evaluation approach depends on the type of model being used.

20.1.1.1 Predictive Models

If the model is a **predictive model**, a **decision tree** can be used to evaluate whether the answers produced by the model are aligned with the initial design.

This evaluation helps us:

- Determine whether the model behaves as expected.
- Identify areas that require adjustments or improvements.

20.1.1.2 Descriptive Models

If the model is **descriptive** (a model in which relationships are being assessed), a **testing set with known outcomes** can be applied.

Using the testing set allows us to:

- Compare the model's output with known results.
- Refine the model as needed.

20.1.2 Statistical Significance Testing

The second phase of model evaluation is **statistical significance testing**.

This type of evaluation is applied to ensure that:

- The data is being properly handled within the model.
- The data is being correctly interpreted by the model.

Statistical significance testing is designed to avoid unnecessary second-guessing when the final answer or result is revealed.

20.2 Case Study: Finding the Optimal Model

To understand the evaluation process, we can examine a case study that applies the **Evaluation** component of the Data Science Methodology.

One way to find the optimal model is through a **diagnostic measure** by tuning one of the parameters used during model building.

In this example, we tune the **relative cost of misclassifying "Yes" and "No" outcomes**.

Four different models were built using four different **relative misclassification costs**.

The results showed that:

- Increasing the model-building parameter increased the **true-positive rate (sensitivity)** for predicting **"Yes."**
- However, this improvement came at the expense of lower accuracy in predicting **"No,"** resulting in an increased **false-positive rate**.

The key question then becomes:

Which model is the best after tuning this parameter?

20.3 Business Considerations

Selecting the optimal model also depends on practical business constraints.

For budgetary reasons:

- The **risk-reducing intervention** could not be applied to most or all **congestive heart failure patients**, many of whom would not have been readmitted anyway.
- On the other hand, if too few high-risk congestive heart failure patients were identified, the intervention would not be as effective in improving patient care.

Therefore, the model needed to balance prediction performance with available resources.

20.4 Receiver Operating Characteristic (ROC) Curve

The optimal model is identified by selecting the model that provides the **maximum separation between the blue ROC curve and the red baseline**.

Based on the evaluation:

- **Model 3**, with a **relative misclassification cost of 4-to-1**, was determined to be the best among the four models.

20.4.1 What is the ROC Curve?

ROC (Receiver Operating Characteristic) Curve is a diagnostic tool used to determine the optimal classification model.

Originally:

- The ROC curve was first developed during **World War II** to detect enemy aircraft on radar.

Today, it is widely used in:

- Machine Learning
- Data Mining

20.4.2 Purpose of the ROC Curve

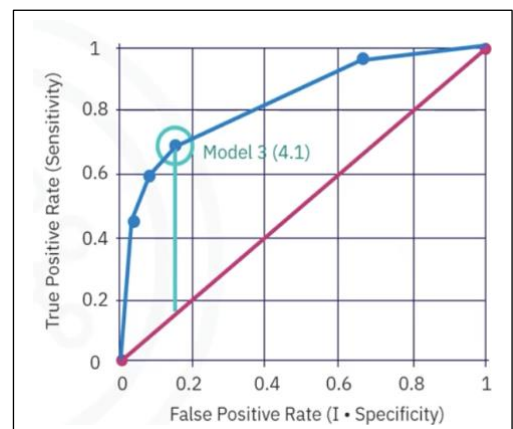
The ROC curve measures how well a **binary classification model** performs by classifying "Yes" and "No" outcomes while varying a discrimination criterion.

In this case, the discrimination criterion is the **relative misclassification cost**.

The ROC curve is created by plotting:

- **True-Positive Rate (Sensitivity)** against
- **False-Positive Rate**

for different values of the relative misclassification cost.



By analyzing this curve, we can select the model that provides the best overall classification performance.

20.5 Key Takeaways

- Model evaluation is performed during model development and before deployment.
- Modeling and evaluation are iterative processes.
- The objective of evaluation is to determine whether the model answers the original question or requires further adjustment.
- Model evaluation consists of two major phases:
 - Diagnostic Measures
 - Statistical Significance Testing

- Diagnostic measures help verify that predictive and descriptive models behave as intended.
- Statistical significance testing ensures that the data is properly handled and interpreted.
- Model tuning involves adjusting parameters such as the **relative misclassification cost**.
- Increasing sensitivity generally increases the false-positive rate, creating a trade-off.
- Business constraints should also be considered when selecting the best model.
- The **ROC (Receiver Operating Characteristic) Curve** is a widely used diagnostic tool for evaluating binary classification models.
- In the case study, **Model 3**, with a **4-to-1 relative misclassification cost**, was identified as the optimal model because it achieved the greatest separation from the ROC baseline.

21 Lesson Summary: From Modeling to Evaluation

We have completed this lesson. At this point, we know:

- The end goal of the **Modeling** stage is for the data model to answer the business question.
- The data modeling process uses a **training data set**. Data scientists test multiple algorithms on the training data to determine whether the variables are required and whether the data supports answering the business question.
- The outcomes of these models are either **descriptive** or **predictive**.

21.1 Modeling to Evaluation

Modeling and evaluation play a pivotal role in shaping analytical outcomes and refining problem-solving strategies.

21.1.1 Data Modeling

- Determines the characteristics and purpose of the modeling process.
- Models can be **descriptive** or **predictive**.
- Models are based on **statistical** or **machine learning** approaches.

21.1.2 Training Set and Calibration

- The **training set** serves as a gauge to calibrate the model.
- Data scientists experiment with different algorithms for **variable selection**.
- The success of the modeling process depends on:
 - Understanding the problem.
 - Selecting an appropriate analytical approach.

21.1.3 Model Evaluation

- Model evaluation is an **iterative process** (repeated and refined as the model improves) and is performed alongside model building.
- Evaluation takes place **before deployment**.
- The purpose is to:
 - Assess the quality of the model.

- Verify that the model aligns with the initial business request.
- Determine whether the model is ready for deployment or requires further adjustments.

21.1.4 Diagnostic Measures

- Diagnostic measures ensure that the model functions as intended.
- **Significance testing** (a statistical method used to determine whether results are meaningful rather than occurring by chance) helps interpret the data.
- The **ROC (Receiver Operating Characteristic) curve** is used to identify the optimal classification model by quantifying model performance.

21.2 Key Takeaways

- The **Evaluation** phase consists of two stages:
 - Diagnostic Measures.
 - Statistical Significance.
- During the Evaluation stage, data scientists and other stakeholders assess the quality of the model and determine whether it answers the initial **Business Understanding** question or whether the data model requires further adjustment.
- The **ROC (Receiver Operating Characteristic) curve** is a valuable diagnostic tool for determining the optimal binary classification model. It measures how effectively a model distinguishes between **yes** and **no** outcomes as the discrimination criterion is varied.

22 Glossary: From Modeling to Evaluation

22.1 Overview

This alphabetized glossary contains many of the terms covered in this topic. These terms are important to recognize when working in the industry and participating in user groups.

22.2 Glossary

Term	Definition
Binary Classification Model	A model that classifies data into two categories, such as yes/no or stop/go outcomes.
Data Compilation	The process of gathering and organizing data required for modeling.
Data Modeling	The stage in the data science methodology where data scientists develop models, either descriptive or predictive , to answer specific questions.
Descriptive Model	A type of model that examines relationships between variables and makes inferences based on observed patterns.
Diagnostic Measure-Based Tuning	The process of fine-tuning the model by adjusting parameters based on diagnostic measures and performance indicators.

Diagnostic Measures	The evaluation of a model's performance to ensure that the model functions as intended.
Discrimination Criterion	A measure used to evaluate the performance of the model in classifying different outcomes.
False-Positive Rate	The rate at which the model incorrectly identifies negative outcomes as positive.
Histogram	A graphical representation of the distribution of a dataset, where the data is divided into intervals or bins, and the height of each bar represents the frequency or count of data points falling within each interval.
Maximum Separation	The point where the ROC curve provides the best discrimination between true-positive and false-positive rates, indicating the most effective model.
Model Evaluation	The process of assessing the quality and relevance of the model before deployment.
Optimal Model	The model that provides the maximum separation between the ROC curve and the baseline, indicating higher accuracy and effectiveness.
Receiver Operating Characteristic (ROC)	Originally developed for military radar, this statistical curve was used by the military to assess the performance of binary classification models.
Relative Misclassification Cost	A parameter in model building used to tune the trade-off between true-positive and false-positive rates.
ROC Curve (Receiver Operating Characteristic Curve)	A diagnostic tool used to determine the performance of the optimal classification model.
Separation	The degree of discrimination achieved by the model in correctly classifying outcomes.
Statistical Significance Testing	An evaluation technique used to verify that data is appropriately handled and interpreted within the model.
True-Positive Rate	The rate at which the model correctly identifies positive outcomes.

Module 3: From Deployment to Feedback and Final Evaluation

23 Deployment

Deployment is the stage where a data science model is put into practical use. While a data science model provides an answer, making that answer relevant and useful for addressing the initial question requires ensuring that stakeholders become familiar with the tool that has been developed.

In a business scenario, stakeholders from different specialties work together to make deployment successful. These stakeholders may include:

- Solution Owner
- Marketing Team
- Application Developers
- IT Administration

Once the model has been evaluated and we are confident that it performs as expected, it is deployed and put to the ultimate test.

Depending on the purpose of the model, deployment may involve:

- Rolling out the model to a limited group of users.
- Deploying it in a test environment.
- Building confidence before applying the solution across the entire organization.

23.1 Case Study: Deployment for Reducing Hospital Readmission Risk

In preparation for solution deployment, the next step was to assimilate the knowledge for the business group responsible for designing and managing the intervention program to reduce readmission risk.

In this scenario, the business team translated the model results so that the clinical staff could understand how to:

- Identify high-risk patients.
- Design suitable intervention actions.

The goal was to reduce the likelihood that these patients would be readmitted within **30 days after discharge**.

23.2 Business Requirements

During the business requirements stage, the Intervention Program Director and her team requested an application that would provide:

- Automated, near real-time risk assessments of congestive heart failure.
- An interface that was easy for clinical staff to use.
- A browser-based application that could run on a tablet carried by each staff member.

23.3 Patient Risk Assessment Process

Patient data was generated throughout the hospital stay.

The deployment process included the following workflow:

- Patient data was automatically prepared in the format required by the model.
- Each patient was scored near the time of discharge.
- Clinicians received the most up-to-date risk assessment for every patient.
- The risk assessment helped clinicians determine which patients should receive intervention after discharge.

23.4 Training and Operational Support

As part of solution deployment, the Intervention Team was responsible for developing and delivering training for the clinical staff.

In addition, processes for tracking and monitoring patients receiving the intervention had to be developed in collaboration with:

- IT Developers
- Database Administrators

These processes enabled the results to proceed through the feedback stage so that the model could be refined over time.

23.5 Example: Cognos Application

A solution was deployed through a **Cognos application**.

In this case study, the objective was to assess the hospitalization risk for patients with **juvenile diabetes**.

Like the congestive heart failure use case, this application used **Decision Tree Classification** to create a risk model that served as the foundation of the application.

The application included the following capabilities:

- A nationwide map providing an overview of hospitalization risk.
- Interactive analysis of predicted risk based on various patient conditions and other characteristics.

- The **Cognos application** demonstrated how **Decision Tree Classification** could be used to predict hospitalization risk for patients with **juvenile diabetes** through interactive reports and patient-specific summaries.
- The feedback collected after deployment supports continuous refinement and improvement of the model over time.

24 Feedback

24.1 Introduction

Once in play, feedback from the users will help to refine the model and assess it for performance and impact.

The value of the model will be dependent on successfully incorporating feedback and making adjustments for as long as the solution is required.

Throughout the Data Science Methodology, each step sets the stage for the next. Making the methodology cyclical ensures refinement at each stage in the process.

The feedback process is rooted in the notion that **the more we know, the more that we'll want to know**. That's the way John Rollins sees it, and hopefully we do too.

24.2 Purpose of the Feedback Stage

Once the model is evaluated and the data scientist is confident it will work, it is deployed and put to the ultimate test: actual, real-time use in the field.

To understand how the Feedback stage is applied, we revisit the case study and examine the planned activities for this phase.

24.3 Feedback Stage Plan

The plan for the Feedback stage included the following steps:

- The review process would be defined and put into place, with overall responsibility for measuring the results of a **"flying to risk" model** of the **congestive heart failure risk population**.
- Clinical management executives would have overall responsibility for the review process.
- Congestive heart failure patients receiving intervention would be tracked, and their re-admission outcomes would be recorded.
- The intervention would then be measured to determine how effective it was in reducing re-admissions.

24.4 Measuring Model Effectiveness

For ethical reasons, congestive heart failure patients would not be split into controlled and treatment groups.

Instead, readmission rates would be compared before and after the implementation of the model to measure its impact.

After the Deployment and Feedback stages, the impact of the intervention program on readmission rates would be reviewed after the first year of its implementation.

24.5 Model Refinement

The model would then be refined based on:

- All of the data compiled after model implementation.
- The knowledge gained throughout the Deployment and Feedback stages.

Other refinements included:

- Incorporating information about participation in the intervention program.
- Possibly refining the model to incorporate detailed pharmaceutical data.

If we recall, data collection was initially deferred because the pharmaceutical data was not readily available at the time.

However, after feedback and practical experience with the model, it might be determined that adding that data could be worth the investment of effort and time.

We also have to allow for the possibility that other refinements might present themselves during the Feedback stage.

24.6 Refinement of Intervention Actions

The intervention actions and processes would also be reviewed and very likely refined based on the experience and knowledge gained through the initial Deployment and Feedback stages.

Finally, the refined model and intervention actions would be redeployed, with the feedback process continued throughout the life of the intervention program.

24.7 Key Takeaways

- Feedback helps refine the deployed model and assess its performance and impact.
- The Data Science Methodology is cyclical, allowing continuous improvement at every stage.
- The review process measures the effectiveness of the deployed solution.
- Clinical management executives oversee the review process in the case study.
- Patient outcomes are tracked to evaluate intervention effectiveness.
- Ethical considerations prevented the use of controlled and treatment groups.
- Model effectiveness is measured by comparing readmission rates before and after implementation.
- The model is refined using newly collected data and operational experience.
- Additional data sources, such as pharmaceutical data, may be incorporated if they prove valuable.

- Intervention actions and processes are also refined based on feedback.
- The refined model is redeployed, and the feedback cycle continues throughout the life of the intervention program.

25 The Role of Storytelling in the Life of a Data Analyst

Storytelling plays a vital role in the life of a data analyst. Its importance cannot be overstated because it helps us communicate insights effectively, influence decisions, and make data meaningful for different audiences.

25.1 Why Storytelling Matters

Humans naturally understand the world through stories. If we are trying to convince anyone to take action based on data, the first thing we have to do is tell a **clear, concise, and compelling story**.

Storytelling is also useful while working with a dataset. Developing a story around the data helps us better understand the underlying dataset and what it is doing.

25.2 Balancing Simplicity and Complexity

There is always a balance between:

- Telling a clear, coherent, and simple story.
- Making sure we convey all the complexities that exist within the data.

Finding this balance can be challenging, but it is a critical skill for every data analyst.

25.3 The Importance of Communication

The art of storytelling is significant in the life of a data analyst. It does not matter how much valuable or wonderful information we have discovered. If we cannot communicate it effectively to our audience, whether they are:

- Consumers
- Directors
- Executive-level decision-makers

then the value of our analysis is lost.

We must find an effective way to communicate insights, and this is usually best achieved through visualizations or storytelling so that the audience understands how the information can be useful.

25.4 Storytelling as the Final Step

Storytelling is an essential skill set. It is like **the last mile in delivery** (the final and most important step in successfully delivering value to the audience).

Many people can learn the technical side of data analysis through a relatively short period of training. However, the ability to:

- Extract value from data.
- Communicate that value effectively.

is in short supply.

If we are thinking about a long-term career, knowing how to tell a compelling story with data is extremely important.

25.5 Storytelling Helps Drive Action

Storytelling is absolutely crucial to data and analytics because it is how we convey our message.

Anyone can present:

- Numbers
- KPIs (Key Performance Indicators)
- Statistics

However, if we do not have a story around those numbers or a compelling reason to act, what we present is unlikely to resonate with our audience.

25.6 Stanford Study on Storytelling

A study at **Stanford** examined how people remembered presentations.

In the study:

- People presented their pitches.
- Their pitches included KPIs, numbers, and statistics.
- They also told a story alongside the data.
- Audience members were later quizzed about what they remembered from each presentation.

The results showed that it was **the stories** that people remembered.

Although the stories still contained facts and figures, presenting the information within a story made it much more memorable.

25.7 Creating an Emotional Connection

Having an emotional connection to:

- The story
- The understanding
- The data

is how we encourage people to take the actions that we want and need them to take.

Storytelling transforms data from a collection of numbers into meaningful insights that inform decisions, inspire confidence, and motivate action.

26 Data Science Methodology 101 Course Summary

We have reached the end of the **Data Science Methodology 101** course. Throughout this course, we learned how to think like a data scientist by following a structured methodology for solving real-world data science problems.

26.1 What We Learned

We learned the complete process of tackling a data science problem, including:

- Forming a concrete business or research problem.
- Collecting and analyzing data.
- Building a model.
- Understanding and using feedback after model deployment.

We also learned methodical ways of moving from a problem to an appropriate approach, including:

- Understanding the question.
- Understanding the business goals and objectives.
- Selecting the most effective analytical approach to answer the question and solve the problem.

In addition, we learned methodical ways of working with data by:

- Determining the data requirements.
- Collecting the appropriate data.
- Understanding the data.
- Preparing the data for modeling.

We also learned how to model the data by selecting the appropriate analytical approach based on the data requirements and the problem we were trying to solve.

Once the analytical approach was selected, we learned the steps involved in:

- Evaluating the model.
- Deploying the model.
- Collecting feedback after deployment.
- Using constructive feedback to improve the model.

26.2 The Iterative Nature of the Data Science Methodology

An important concept introduced in this course is that the stages of the Data Science Methodology are **iterative** (repeated continuously to improve the solution).

This means that a model can always be improved for as long as the solution is needed, regardless of whether the improvements come from:

- Constructive feedback.
- Newly available data sources.

Continuous improvement ensures that the model remains effective and relevant over time.

26.3 Applying the Methodology in a Real-World Case Study

Using a real case study, we learned how the Data Science Methodology can be applied in context to successfully achieve the goals established during the **Business Requirements** stage.

We also observed how the methodology added value to business units by incorporating data science practices into their daily analysis and reporting functions.

The success of the pilot program presented in the case study was demonstrated by the fact that physicians were able to deliver better patient care by using new tools that incorporated timely, data-driven information into patient care decisions.

26.4 The True Purpose of a Methodology

In summary, we learned that the true purpose of a methodology is to:

- Explain how to approach a problem.
- Work with data to support problem-solving.
- Develop a solution that addresses the root problem.

By answering **10 simple questions** methodically, we learned that a methodology can help solve not only data science problems but also many other types of problems.

26.5 Final Takeaway

Success in the field of Data Science depends on our ability to:

- Apply the right tools.
- Apply them at the right time.
- Follow the correct sequence.
- Solve the right problem.

As **John Rollins** stated, this is the proper way to approach Data Science.

27 Module 3 Summary: From Deployment to Feedback

We have completed this lesson. At this point in the course, we know:

- Stakeholders, including the solution owner, marketing staff, application developers, and IT administration, evaluate the model and contribute feedback.
- During the **Deployment** stage, data scientists release the data model to a targeted group of stakeholders.
- Stakeholders and user feedback help assess the model's performance and impact during the **Feedback** stage.
- The model's value depends on iteration; that is, how successfully the data model incorporates user feedback.

27.1 From Deployment to Feedback

27.1.1 Stakeholder Engagement

Making the answer relevant and useful involves engaging stakeholders. Stakeholders include:

- Solution owners
- Marketing staff
- IT administrators
- Application developers

The diverse specialties of these stakeholders help ensure the model's applicability and usefulness in real-world scenarios.

27.1.2 Deployment and Feedback

After the model has been evaluated and the data scientists are confident in its performance, it is deployed to a targeted group of stakeholders for real-time functionality testing.

User feedback is critical for:

- Refining the model
- Evaluating model performance
- Enhancing the model's effectiveness

27.1.3 Iterative Process

The Data Science Methodology follows a cyclical process that continuously refines each stage.

Key points include:

- Feedback drives continuous learning and enhancement.
- Refinement relies on post-implementation data and newly acquired knowledge.
- Additional improvements can be identified and implemented based on stakeholder and user feedback.

27.1.4 Improvement and Redeployment

Feedback insights are integrated to refine both the data model and the associated interventions.

The refined model and actions are then redeployed while continuing to collect stakeholder and user feedback.

This continuous improvement loop:

- Supports ongoing model enhancement.
- Enables continuous assessment of the model's impact.
- Incorporates ethical considerations throughout the refinement and redeployment process to ensure responsible and effective deployment.

28 Module 3 Glossary: From Deployment to Feedback

This topic provides an alphabetically organized glossary of important terms related to the deployment, evaluation, and continuous improvement of data science solutions. These terms are essential for understanding industry practices and participating in professional communities.

28.1 Glossary

Term	Definition
Browser-based application	An application that users access through a web browser, typically on a tablet or other mobile device, to provide easy access to the model's insights.
Cyclical methodology	An iterative approach to the data science process, where each stage informs and refines the subsequent stages.
Data collection refinement	The process of obtaining additional data elements or information to improve the model's performance.
Data science model	The result of data analysis and modeling that provides answers to specific questions or problems.
Feedback	The process of obtaining input and comments from users and stakeholders to refine and improve the data science model.
Model refinement	The process of adjusting and improving the data science model based on user feedback and real-world performance.
Redeployment	The process of implementing a refined model and intervention actions after incorporating feedback and improvements.
Review process	The systematic assessment and evaluation of the data science model's performance and impact.

Solution deployment	The process of implementing and integrating the data science model into the business or organizational workflow.
Solution owner	The individual or team responsible for overseeing the deployment and management of the data science solution.
Stakeholders	Individuals or groups with a vested interest in the data science model's outcome and its practical application, such as solution owners, marketing, application developers, and IT administration.
Storytelling	The art of conveying our message or ideas through a narrative structure that engages, entertains, and resonates with the audience.
Test environment	A controlled setting where the data science model is evaluated and refined before full-scale implementation.

Module 4: Final Project and Assessment

29 Introduction to CRISP-DM

29.1 Learning Objectives

After studying this topic, we will be able to:

- Define **CRISP-DM (Cross-Industry Standard Process for Data Mining)**.
- List and describe the six stages of the **CRISP-DM** model.
- Explain what happens after the final stage of the **CRISP-DM** process.

29.2 What is CRISP-DM?

CRISP-DM (Cross-Industry Standard Process for Data Mining) is an industry-proven methodology for guiding data mining efforts.

It is:

- An **iterative** (repeated and continuously improving) data mining model.
- A comprehensive methodology for data mining projects.
- A structured approach that guides **data-driven decision-making**.
- A process model that provides high-level insights into the **data mining lifecycle**.

As a data methodology, CRISP-DM includes:

- Six data mining stages.
- Descriptions of each stage.
- Explanations of the relationships between tasks and stages.

Like other data science methodologies, CRISP-DM requires:

- Flexibility at every stage.
- Continuous communication with peers, management, and stakeholders to keep the project on track.

Since CRISP-DM is iterative, after completing any stage, data scientists may need to revisit an earlier stage and make necessary changes.

29.3 The Six Stages of CRISP-DM

29.3.1 Business Understanding

The **Business Understanding** stage is the most important stage because it sets and outlines the intentions of the data analysis project.

This stage is common to both:

- John Rolland's (also referred to in the transcript as John Rowland's) Data Science Methodology.
- CRISP-DM methodology.

This stage requires:

- Communication.
- Clarity.
- Collaboration with stakeholders.

Its purpose is to overcome:

- Differing stakeholder objectives.
- Biases.
- Information-related modalities (different ways stakeholders interpret and communicate information).

Without a clear, concise, and complete understanding of the business problem and project goals, the project can waste significant time and resources.

29.3.2 Data Understanding

CRISP-DM combines the following stages from John Rolland's methodology into a single **Data Understanding** stage:

- Data requirements.
- Data collection.
- Data understanding.

During this stage, data scientists:

- Decide on appropriate data sources.

- Acquire the required data.

29.3.3 Data Preparation

During the **Data Preparation** stage, data scientists:

- Transform the collected data into a usable data subset.
- Determine whether additional data is required.
- Select an appropriate dataset.
- Address questionable, missing, or ambiguous data values.

Data Preparation is common to both:

- Foundational Data Methodology.
- CRISP-DM.

29.3.4 Modeling

The **Modeling** stage fulfills the primary purpose of data mining.

During this stage, data scientists:

- Create data models.
- Reveal patterns and structures within the data.
- Generate knowledge and insights that address the stated business problem and goals.
- Select models based on subsets of the data.
- Adjust models whenever necessary.

Model selection is both an art and a science.

Both Foundational Methodology and CRISP-DM focus on creating knowledge and information that has meaning and practical utility.

29.3.5 Evaluation

During the **Evaluation** stage, data scientists test the selected model.

Typically, they:

- Prepare a pre-selected test.
- Run the trained model on the test data.
- Allow the test platform to treat the data as new data.
- Assess the effectiveness of the model.

The testing results determine:

- The model's efficacy (effectiveness).
- The model's role in the final deployment stage.

29.3.6 Deployment

During the **Deployment** stage:

- Data scientists and stakeholders use the model on new data that is outside the scope of the original dataset.
- New interactions may reveal new variables.
- These interactions may indicate the need for:
 - A different dataset.
 - A different model.

29.4 The Iterative Nature of CRISP-DM

CRISP-DM is an **iterative and cyclical** methodology.

Deployment results may initiate revisions to:

- Business needs and actions.
- The model.
- The data.
- Any combination of these.

Therefore, after any stage, it may be necessary to return to an earlier stage to improve the overall solution.

29.5 What Happens After the Final Stage?

After completing all six stages:

- We conduct another **Business Understanding** meeting with the stakeholders to discuss the results.
- In CRISP-DM, this stage is **not explicitly named**.
- However, in **John Rolland's (John Rowland's) Data Science Methodology**, this stage is explicitly called the **Feedback Stage**.

The CRISP-DM process continues until:

- Stakeholders,
- Management, and
- We

agree that the data model and its analysis provide the answers needed to resolve the business problems and achieve the business goals.

29.6 Summary

CRISP-DM stands for **Cross-Industry Standard Process for Data Mining**.

It is an industry-proven, iterative methodology that provides a structured approach to data mining and data-driven decision-making.

The six stages of CRISP-DM are:

- Business Understanding
- Data Understanding
- Data Preparation
- Modeling
- Evaluation
- Deployment

The process is iterative, meaning that after any stage—including deployment—we may revisit previous stages to improve the project. The CRISP-DM process continues until stakeholders, management, and we agree that the data model and its analysis successfully answer the business questions and support the organization's goals.

30 Final Assignment Overview

30.1 Overview

In this topic, we will demonstrate our understanding of the **Data Science Methodology** by applying the **CRISP-DM (Cross-Industry Standard Process for Data Mining)** framework to a real-world business problem.

During this final project, we will demonstrate our knowledge of the **CRISP-DM data methodology**.

30.2 Project Roles

For this project, we will step into two different roles:

30.2.1 The Client

Imagine ourselves as the **business stakeholder**. Our responsibility is to define a **real-world business problem** in one of the following domains:

- Emails
- Hospitals
- Credit Cards

We should think about the challenges faced in the selected domain and phrase the problem as a **data-driven question** that could be answered through data analysis.

30.2.2 The Data Scientist

Once the business problem is defined, we will switch perspectives and take on the role of the **Data Scientist**.

In this role, we will apply each stage of the **CRISP-DM** methodology to explain how data can be used to solve the identified business problem.

For each stage of CRISP-DM, we should:

- Describe the approach we would take.
- Explain the reasoning behind our decisions.
- Demonstrate how the business need can be transformed into a data-driven solution.

30.3 Summary

By the end of this assignment, we will have hands-on experience applying the **CRISP-DM** methodology to a real-world business challenge.

This assignment will strengthen our ability to connect **business needs** with **data-driven solutions**, which is an essential skill for every **Data Scientist**.

31 Final Project: CRISP-DM Data Science Methodology

31.1 Selected Topic

“Credit Card Fraud Detection”

31.2 Business Understanding

31.2.1 Problem Statement

Credit card companies lose millions of dollars each year due to fraudulent transactions. Detecting fraud quickly while minimizing false alarms is a major challenge.

31.2.2 Data Question

Can transaction data be used to accurately identify whether a credit card transaction is fraudulent accurately?

31.2.3 Objective

Develop a predictive model to detect fraudulent credit card transactions, helping financial institutions reduce losses and improve customer security.

31.3 Analytic Approach

Since the goal is to predict whether a transaction is fraudulent or legitimate, the appropriate analytic approach is **classification**, a supervised machine learning technique.

The process would include:

- Exploring historical transaction data.
- Identifying patterns that distinguish fraudulent transactions.

- Training a classification model.
- Using the model to predict fraud in new transactions.

31.4 Data Requirements

The following data would be required:

- Transaction ID
- Transaction amount
- Transaction date and time
- Merchant information
- Customer ID
- Payment method
- Transaction location
- Device or IP address (if available)
- Whether the transaction was fraudulent (target variable)

The data should contain both fraudulent and legitimate transactions for model training.

31.5 Data Collection Plan

The required data can be collected from:

- Historical transaction records from the bank.
- Fraud investigation reports.
- Customer account databases.
- Payment processing systems.

Before analysis, we should:

- Remove personal identifying information.
- Ensure compliance with privacy regulations.
- Combine data from multiple sources into one dataset.

31.6 Data Understanding and Preparation

31.6.1 Data Understanding

We should:

- Examine the dataset structure.
- Check for missing values.
- Identify duplicate records.
- Explore fraud frequency.
- Analyze transaction distributions.

31.6.2 Data Preparation

We should:

- Remove duplicates.
- Handle missing values.
- Convert categorical variables into numerical format.
- Normalize transaction amounts if necessary.
- Split the dataset into training and testing sets.

31.7 Modeling Approach

A **Random Forest Classifier** would be used because it:

- Performs well on classification problems.
- Handles large datasets effectively.
- Works well with many input variables.
- Reduces overfitting compared to a single decision tree.
- Provides high prediction accuracy.

Other possible models include:

- Logistic Regression
- Decision Trees
- XGBoost

31.8 Evaluation Plan

The model would be evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC Score
- Confusion Matrix

Because fraud detection datasets are often imbalanced, **precision** and **recall** are especially important. A successful model should correctly identify fraudulent transactions while minimizing false positives.

31.9 CRISP-DM Data Requirements

The **Data Requirements** phase identifies:

- What data is needed.
- The type and format of the data.
- The sources of the data.
- The quality requirements for the data.

This phase ensures that the correct information is available before data collection begins.

32 Review What We Learned

We have learned many important concepts related to the data science methodology. The following are **14 key, high-level takeaways** to remember.

32.1 Foundational Methodology

- **Foundational Methodology** is a **cyclical, iterative data science methodology** developed by **John Rollins**.
- It consists of **10 stages**, beginning with **Business Understanding** and ending with **Feedback**.

32.2 CRISP-DM Methodology

- **CRISP-DM** is an **open-source data science methodology**.
- It combines several data-related methodology stages into a single stage and **omits the Feedback stage**, resulting in a **six-stage data methodology**.

32.3 Business Understanding

- The primary goal of the **Business Understanding** stage is to:
 - Understand the business problem.
 - Determine the data needed to answer the core business question.

32.4 Analytic Approach

- During the **Analytic Approach** stage, we choose:
 - A **descriptive, diagnostic, predictive, or prescriptive** analytical approach.
 - Whether to use **machine learning techniques**.

32.5 Data Requirements

- During the **Data Requirements** stage, data scientists identify:
 - The correct and necessary **data content**.
 - Required **data formats**.
 - Appropriate **data sources**.
- These requirements depend on the selected analytical approach.

32.6 Data Collection

- During the **Data Collection** stage, expert data scientists:
 - Revise the initial data requirements.
 - Make critical decisions regarding the **quantity** and **quality** of the data.
 - Apply **descriptive statistics** and **visualization techniques** to:
 - Assess the data content.
 - Evaluate data quality.
 - Gain initial insights from the collected data.

- Identify gaps in the data.
- Determine whether new data is required or whether existing data should be substituted.

32.7 Data Understanding

- The **Data Understanding** stage encompasses all activities related to constructing the dataset.
- This stage answers the question:

Does the collected data represent the data needed to solve the business problem?

- Data scientists may use:
 - **Descriptive statistics**
 - **Predictive statistics**
 - Or both.
- Common techniques and statistics include:
 - **Hurst** (Hurst exponent, a statistical measure used to evaluate long-term memory or persistence in time-series data)
 - **Univariates** (analysis of one variable at a time)
 - Mean
 - Median
 - Minimum
 - Maximum
 - Standard deviation
 - Pairwise correlation
 - Histograms

32.8 Data Preparation

- During the **Data Preparation** stage, data scientists:
 - Address missing or invalid values.
 - Remove duplicate records.
 - Validate that the data is properly formatted.
- Key techniques applied during this stage include:
 - **Feature engineering** (creating or transforming variables to improve model performance)
 - **Text analysis** (extracting meaningful information from textual data)
- These techniques help validate and analyze the data before modeling.

32.9 Modeling

- The end goal of the **Modeling** stage is to ensure that the data model answers the business question.
- During this stage, data scientists:
 - Use a **training dataset**.
 - Test multiple algorithms on the training data.
 - Determine whether the variables are required.
 - Evaluate whether the data supports answering the business question.

- The resulting models are either:
 - **Descriptive**, or
 - **Predictive**.

32.10 Evaluation

- The **Evaluation** stage consists of two phases:
 - **Diagnostic measures phase**
 - **Statistical significance phase**
- During this stage, data scientists and other stakeholders:
 - Assess the quality of the model.
 - Determine whether the model answers the original **Business Understanding** question.
 - Decide whether the model requires further adjustment.

32.11 Deployment

- During the **Deployment** stage, data scientists release the data model to a targeted group of stakeholders, including:
 - Solution owners
 - Marketing staff
 - Application developers
 - IT administration

32.12 Feedback

- During the **Feedback** stage:
 - Stakeholders and users evaluate the model.
 - They provide feedback to assess the model's performance.
- The value of a data model depends on its ability to **iterate** (continuously improve by incorporating user feedback).